Aridity and herbivory shape demographic benefits of fungal endophytes in cool-season grasses

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Abstract

Plant–microbe symbioses are generally expected to enhance host resilience to environmental stress, yet whether effects differ across abiotic versus biotic stress dimensions remains unclear. We tested how aridity and herbivory shape fitness effects of vertically transmitted fungal endophytes on three grass hosts using herbivore exclusion experiments along a precipitation gradient in the south-central United States. Contrary to theoretical predictions, endophyte-symbiotic hosts showed disproportionately stronger declines in demographic performance with increasing aridity than endophyte-free plants, suggesting abiotic stress exacerbated symbiont costs. Endophyte symbiosis reduced herbivore damage for two host species, consistent with protective benefits, but exclusion effects were weak and idiosyncratic relative to climate stress. A greenhouse experiment using soils from field sites showed endophyte effects were insensitive to soil origin, supporting a causal role of climate in driving site-level variation. Together, these results challenge the expectation that environmental stress enhances symbiosis benefits and suggest limited buffering potential under increasing aridity.

Introduction

Mutualistic symbioses are widespread and ecologically important, often mediating stress tolerance and defense against natural enemies for host organisms. Like mutualisms more broadly, host-symbiont interactions may be strongly context-dependent, with the direction and strength of outcomes shaped by the environment in which they occur (Bronstein, 1994; Hoang et al., 2024; Hoeksema and Bruna, 2015; Rudgers et al., 2020). The stress-gradient hypothesis (SGH) is a classic conceptual framework for variation in interaction outcomes, predicting that the frequency of positive interactions increases under stressful conditions (Bertness and Callaway, 1994; Holmgren and Scheffer, 2010; Louthan et al., 2015; Maestre et al., 2009). For example, microbial symbionts that are costly to maintain under benign conditions may benefit host plants under drought stress (Giauque et al., 2019). Stress can take many forms, including abiotic and biotic factors, and it remains unclear how predictions of the SGH for host-symbiont interactions play out over different dimensions of environmental stress.

Under biotic stress, such as competition or attack by natural enemies, symbionts can mediate defensive traits across diverse taxa (Atala et al., 2022; Bastias et al., 2017; Flórez et al., 2015; Haine, 2008; Vega, 2008). Similarly, under abiotic stress, including drought, nutrient limitation, and thermal extremes, symbiotic partners often enhance host tolerance through improved resource acquisition, physiological regulation, and stress-responsive metabolism (Afkhami and Rudgers, 2008; Cesar et al., 2024; Clay and Schardl, 2002). In plants specifically, microbial symbionts such as mycorrhizae, rhizobia, and endophytic bacteria and fungi can promote water and nutrient uptake, nitrogen fixation, and resistance to herbivores, allowing hosts to maintain growth and function under harsh abiotic or biotic conditions (Amijee et al., 1989; Friesen et al., 2011; van Der Heijden et al., 2015). Many plant symbionts also stimulate the production of antioxidant enzymes, reducing oxidative damage caused by stressors such as drought, salinity, and heat (Ruiz-lazona et al., 1996). By improving resource acquisition, modulating plant physiology, and enhancing stress-responsive metabolism, these symbiotic interactions help host plants buffer against environmental stochasticity and extreme conditions across a wide range of ecosystems (Fowler et al., 2023; Li et al., 2026; Rudgers et al., 2020).

While stress is often thought to enhance the benefits of symbiotic microbes, growing evidence suggests it can also increase the costs of hosting them. In plants, maintaining symbionts requires that hosts allocate valuable carbon and nutrients; for example, mycorrhizal fungi can demand up to 20% of a host's photosynthetically fixed carbon (Soudzilovskaia et al., 2015; Thirkell et al., 2020). Under abiotic stress, this investment may outweigh benefits, reducing survival, growth, reproduction, or competitive ability (Bronstein, 2001; Johnson et al., 1997; Klironomos, 2003). Analogously, in coral bleaching, heat stress disrupts the mutualism between corals and algal symbionts (Marangon et al., 2025; Vidal-Dupiol et al., 2009). However, across systems, the frequency of such symbiosis breakdown and the environmental or biological thresholds that trigger them remain poorly understood. Quantifying these limits is essential for predicting when stress will enhance mutualistic outcomes and when it will impose demographic costs on hosts.

Despite growing interest in the role of symbiosis in shaping host demography, few studies have tested how abiotic and biotic stressors synergistically influence the demographic outcomes of symbiosis. It therefore remains unclear how the relative roles of these stressors shape the benefits and costs of symbionts. In a greenhouse study of *Ammophila breviligulata*, herbivory reduced plant performance, but endophyte-symbiotic plants were more resistant to both herbivory and aridity (Emery et al., 2010). Studying the combined effects of abiotic and biotic stressors under natural conditions is particularly important in the context of global change (Louthan et al., 2015), as shifting climate patterns and altered herbivore pressures may change the balance of costs and benefits in plant–microbe interactions, potentially influencing species distributions and ecosystem function (Reynolds et al., 2003). Understanding how abiotic stress, biotic stress, and microbial symbiosis interact requires experiments that manipulate all three factors simultaneously, yet such fully factorial studies remain rare outside controlled settings. Here, we address this gap by examining the demographic effects of endophyte symbiosis on three grass host species spanning a natural range-core/range-edge precipitation gradient and in the presence/absence of herbivores.

Symbioses between cool-season grasses and seed-transmitted fungal endophytes (*Epichloë* spp.) provide a powerful experimental model for studying how abiotic and biotic elements of ecological context influence host–symbiont interactions (Clay et al., 2005; Oberhofer et al., 2014; Saikkonen et al., 2010; Schardl et al., 2004). Because vertical transmission links host and symbiont fitness, it is expected to favor the evolution of mutualistic interactions (Gundel et al., 2011, 2024). Environmental conditions, including long-term variation in precipitation, temperature, and herbivore exposure, strongly influence both the prevalence of *Epichloë* infection and the benefits conferred to hosts (Clay et al., 2005; Fowler et al., 2025; Rudgers et al., 2016). For example, grasses hosting endophytes often outperform endophyte-free individuals under water limitation, though the magnitude of these benefits depends on both host and symbiont identity (Decunta et al., 2021). A key benefit of *Epichloë* symbiosis is the production of fungal alkaloids that deter herbivory or reduce herbivore performance, thereby mediating plant–herbivore interactions (Crawford et al., 2010; Saikkonen et al., 2010). Fungal alkaloids may also modify host osmotic balance in ways that promote water use efficiency (Decunta et al., 2021). Endophytes may impose costs on their hosts, including the carbon resources required to sustain the fungus and reduced growth or reproduction when fungal demands exceed host resource supply (Bennett and Groten, 2022). Finally, *Epichloë* endophyte effects can differ across host vital rates, which makes it important to consider multiple vital rates in assessing costs and benefits: for example, symbiosis may not improve survival under stress, but can still enhance growth and reproduction (Rudgers et al., 2012; Yule et al., 2013).

We established experimental gardens of three species of native cool-season grasses across an aridity gradient in the south-central United States to investigate how endophyte symbiosis affects multiple host vital rates under varying abiotic and biotic stress. At each of seven sites along the gradient, we manipulated vertebrate and invertebrate herbivory (allowing herbivore access versus exclusion) to test its interaction with abiotic stress. We asked whether endophyte symbiosis is beneficial for host grasses, whether benefits are amplified under abiotic and/or biotic stress as predicted by the stress-gradient hypothesis (SGH), and how these stressors interact to shape demographic outcomes. We evaluated four alternative hypotheses (Fig. 1):

1. Context-independent mutualism: Endophytes confer consistent positive effects on host demography across environmental conditions (Fig. 1a).
2. Herbivory-dependent effects: Endophyte benefits are greater under herbivore exposure than under herbivore exclusion, independent of the abiotic environment, consistent with a defensive role (Fig. 1b).
3. Climate-dependent effects: Endophyte benefits increase under stressful, arid conditions and weaken or become negative in favorable, mesic environments, independent of herbivore exposure (Fig. 1c).
4. Herbivory- climate-dependent effects: Endophyte effects depend on the joint influence of herbivory and precipitation, with the strongest benefits when both stressors co-occur. We further hypothesized that net benefits may be weaker than predicted by either stressor alone, reflecting the increased physiological cost of maintaining the association under compounded stress (Fig. 1d).

Finally, to supplement the geographically-distributed field experiment and isolate the causal role of climatic factors, we used a greenhouse experiment with one host species to assess the influence of edaphic factors that co-vary with precipitation across experimental sites.

Materials and methods

Study system

Our study focused on three cool-season perennial grass species native to woodland, savanna, and grassland habitats of eastern North America: *Agrostis hyemalis*, *Elymus virginicus*, and *Poa* *autumnalis* (Poaceae, subfamily Pooideae) (Shaw, 2011). *A. hyemalis* is a short-lived perennial that germinates from autumn to late winter and, in our Texas study region, typically flowers between April and June. *E. virginicus* is a larger, longer-lived species that flowers throughout spring and summer. *P. autumnalis* has a similar growth form to *E. virginicus* and flowers in late spring. *Agrostis hyemalis* hosts the endophyte *Epichloë amarillans* (White Jr, 1994), *E. virginicus* hosts the endophyte *Epichloë elymi* (Liu, 2023), and *P. autumnalis* hosts the endophytes *Epichloë PauTG-1* or *E. typhina subsp. poae* (Gundel et al., 2020). Like other *Epichloë* species, these endophytes are primarily transmitted vertically through seeds but are additionally capable of horizontal transmission via sexual stroma on inflorescences or asexual conidia on leaf surfaces (Clay and Schardl, 2002; Gundel et al., 2020; Leuchtmann et al., 2014). In our study, stroma were never observed on *E. virginicus* or *Poa autumnalis* and rarely observed on *A. hyemalis* ($<5\%$ of individuals), suggesting a dominant role of vertical transmission. Herbivory in this system is primarily from deer, generalist insects (aphids, grasshoppers), and small mammalian grazers.

Source populations and symbiont removal

Plants used in our experiment were derived from 11 natural source populations located throughout the ranges of the focal species (Fig. 2, Table S1). Seeds from these source populations (collected from ca. 30 individuals per population) were either heat-treated to remove endophytes or left untreated (control), intended to produce lineages of symbiont-free (S-) and symbiotic (S+) plants from the same genetic background. For *E. virginicus* and *P. autumnalis*, heat-treated seeds were placed in a drying oven at 60°C for five days. For *A. hyemalis*, seeds were heat-treated in a water bath at 60°C for 12 minutes. For all but two populations, the S- plants used in our experiment were reared through one or more generations following heat treatment, which minimized the confounding potential of direct physiological effects of heating seeds (Table S2).

Seeds from control and heat-treated lineages were surface-sterilized with a 10% bleach solution and germinated on water agar plates (18.2 g agar per L deionized water). Seedlings were transferred to potting soil and grown in the Rice University greenhouse. Once these plants were sufficiently large they were vegetatively propagated to help meet target sample sizes, with clonal identities tracked across individuals. After vegetative propagation of *A. hyemalis*, *E. virginicus*, and *P. autumnalis*, the experiment included, respectively, 840, 840, and 720 plants (half S+, half S-) from three, four, and four source populations, with 92.3, 112.0, and 18.0 mean genotypes per source population and 5.47, 1.73, and 12.9 mean clonal replicates per genotype (Table S1). Endophyte status of all plants was verified prior to field transplant using microscopy and immunoblot assays (see Supplementary Methods).

Common garden experimental design and climate data

We established common garden experiments with S+ and S- hosts at seven sites along an aridity gradient from Louisiana to central Texas, US (Fig. 2, Table S3). From west to east, mean annual precipitation increases more than three-fold across these sites while mean annual temperature is relatively consistent, based on climate data collected during the same period as our demographic surveys (Fig. 2d, Fig. S1). For this reason, we focus on precipitation as the primary abiotic factor that varied across sites. This geographic gradient overlaps and exceeds the natural range edges of our focal species, ensuring exposure to realistic climatic and biotic conditions (Fig. 2a, b, c).

To characterize climate at these sites during the study period (2023–2025), we collected monthly temperature and precipitation data from PRISM at 800 m resolution (Hart and Bell, 2015). For each site, we calculated mean temperature (°C) and cumulative precipitation (mm) over the corresponding census year, defined as the period from transplanting to the demographic census.

At each of seven sites along the gradient, we established 24 plots (1.5 m $\times$ 1.5 m), eight per host species (*P. autumnalis* was planted at six sites, excluding the westernmost site (SON) since it is well outside this species' natural range: Fig. 2c). Within each site, each plot was randomly assigned to either herbivore access or herbivore exclusion (Fig. 2e). The herbivory treatment targeted both vertebrate and invertebrate herbivores: exclusion plots had deer fencing on four sides and were sprayed with Sevin insecticide (active ingredient Zeta-Cypermethrin) 2-3 times per growing season following the manufacturer's protocol; herbivore access plots were unsprayed and fenced on two sides to control for any fence effect.

Greenhouse-raised plants were planted into the field experiment in February–March 2023. For each species at each site, we planted a total of 60 S+ and 60 S- plants, half of those experiencing herbivore access and half experiencing herbivore exclusion. Each plot was planted with 15 individuals, a mix of S+ and S- plants to create variation in initial endophyte prevalence (not further considered here). Clonal replicates were arranged such that all plots had the same genetic diversity (number of unique genotypes) for each species. Plots at the two westernmost sites (KER and SON) were destroyed by deer and subsequently replanted in 2024 with stronger fencing.

Demographic data collection

We collected demographic data, including survival, growth, and reproduction, during the flowering season of each species: *Agrostis hyemalis* and *Poa autumnalis* were censused in May and *Elymus virginicus* was censused in June of 2023, 2024 and 2025. For each individual, survival was recorded as alive or dead and, for surviving plants, size was measured as the count of tillers. For surviving plants, growth was quantified as the log ratio of tiller count between consecutive censuses (i.e., $\log(\text{tiller}_{t+1}/\text{tiller}_t)$). We recorded the number of inflorescences per plant for all species and counted the number of spikelets on up to three inflorescences from reproducing individuals of *Poa autumnalis* and *Elymus virginicus*. Spikelet counts were not recorded for *Agrostis hyemalis* because its inflorescences contain numerous very small spikelets that are difficult to count reliably in the field. In *A. hyemalis*, one spikelet yields one seed, while in *P. autumnalis* and *E. virginicus* a single spikelet can produce 2-6 seeds.

We measured herbivore damage to assess the effects of herbivore exclusion treatment on realized herbivory. For each individual at each census, we counted the number of tillers with any signs of herbivore (leaf chewing or clipped tillers) and quantified the proportion of total tillers with herbivore damage. We also binned plants as damaged/undamaged based on the presence/absence of any damaged tillers, and tallied the proportion of damaged plants by plot, site, and herbivore treatment.

Statistical modeling

To assess how abiotic and biotic stress modify the demographic effects of endophyte symbiosis, we fit generalized linear mixed models for each vital rate response in a hierarchical Bayesian statistical framework. All models shared the same hierarchical structure for their expected values and were applied to survival, growth, and fertility. Survival was modeled with a Bernoulli distribution, growth with a Gaussian distribution, inflorescence count with a negative binomial hurdle model, and spikelet count with a negative binomial distribution. For inflorescence count, we compared hurdle and zero-inflated negative binomial mixture models. Predictive performance was nearly identical between models based on leave-one-out cross-validation (ΔELPD = 0.2, SE = 3.1), and we retained the hurdle model because it better reflected the biological distinction between flowering occurrence and inflorescence production. All models included species-specific fixed effects for symbiosis (S+ or S-), site- and year-specific precipitation (cumulative rainfall [mm] over 12 months preceding the census), and herbivory (herbivore access/exclusion), including all two- and three-way interactions. We considered models with linear and quadratic forms for the influence of precipitation, to account for the possibility of non-monotonic responses. Although we evaluated quadratic models, model selection criteria (LOOCV) showed that they provided minimal improvement over first-order regression models (Table S4). Therefore, all subsequent analyses used the first-order (linear) effect of precipitation (Eq. 1), which captures the observed patterns while maintaining model simplicity. For each vital rate model, the expected value for the i^{th} observation (μ_i) was given by:

$$
\begin{aligned}
\mu_i = & \beta_0[Sp_i] + \beta_P[Sp_i] * precip_i + \beta_E[Sp_i] * endo_i + \beta_H[Sp_i] * herb_i \\
& + \beta_{EP}[Sp_i] * endo_i * precip_i + \beta_{HP}[Sp_i] * herb_i * precip_i \\
& + \beta_{HE}[Sp_i] * herb_i * endo_i + \beta_{HEP}[Sp_i] endo_i * herb_i * precip_i \\
& + \phi_{\text{site}[i],Sp_i} + \omega_{\text{source}[i]} + \rho_{\text{plot}[i]} \\
& \quad \phi_{\text{site}[i],Sp_i} \sim N(0, \sigma^2_{site}) \\
& \quad \rho_{\text{plot}[i]} \sim N(0, \sigma^2_{plot}) \\
& \quad \omega_{\text{source}[i]} \sim N(0, \sigma^2_{source})
\end{aligned}
\tag{1}
$$

Here, $precip_i$ is realized annual precipitation, $endo_i$ indicates endophyte presence ($endo_i = 1$) or absence ($endo_i = 0$), and $herb_i$ indicates fencing treatment: herbivore access ($herb_i = 0$) or exclusion ($herb_i = 1$). The model includes random effects accounting for background variation at multiple hierarchical levels: $\phi_{\text{site}[i],Sp_i}$ captures site-to-site heterogeneity that is not explained by precipitation, $\rho_{\text{plot}[i]}$ captures plot-to-plot variation within sites, and $\omega_{\text{source}[i]}$ captures variation associated with the provenance of transplants (the latter two are unique to species and therefore not indexed by species). Random effects for site, plot, and source are host species-specific but the variance parameters (σ_{site}, σ_{source}, σ_{plot}) are shared across species; this assumes, for example, that site-to-site heterogeneity is of a similar magnitude across species, even as the model allows for "better" and "worse" sites to differ by species. This hierarchical structure allows the model to "borrow strength" across species (Gelman and Hill, 2007), improving estimates of fixed effects while still capturing species-specific responses to the environmental gradient.

All models were implemented in R (R Core Team, 2023) using Stan via the `rstan` interface (Stan Development Team, 2024). Models were fit using four Markov chains with 3,000 iterations each, including a burn-in period of 1,000 warm-up iterations, which was sufficient to ensure convergence. Fixed effects were assigned weakly informative normal priors. Random effects were modeled with zero-mean normal distributions with diffuse variance priors. Full details on priors and model settings are provided in the accompanying code. We confirmed that models effectively captured key aspects of the grass demography data, including means, standard deviations, and quantiles using posterior predictive checks (Fig. S5).

We used posterior parameter estimates to visualize predictions for how effects of symbiosis varied across abiotic and biotic contexts. For each herbivory treatment (Access or Exclusion) spanning the precipitation gradient, we computed posterior predictions for the vital rates of S+ and S- hosts (Eq. 1). From these predictions, we calculated the difference for each posterior sample ($\Delta = \mu^{+} - \mu^{-}$) to estimate median effects and 90% credible intervals.

We then computed the posterior probability that $\Delta > 0$, which quantifies the likelihood that endophytes enhance performance, given our uncertainty. For instance, a high posterior probability (e.g., $\Pr(\Delta > 0) > 0.9$) indicates strong evidence of an endophyte benefit, whereas a low posterior probability (e.g., $\Pr(\Delta > 0) < 0.1$) indicates strong evidence of an endophyte cost. Intermediate probabilities (approximately 0.7–0.9 or 0.1–0.3) indicate moderate evidence for positive or negative effects, respectively, while probabilities near 0.5 indicate little or no effect.

Greenhouse experiment

We used site variation to capture an aridity gradient, but other abiotic factors, especially edaphic conditions, covary across sites. For example, the two westernmost sites (KER and SON; Fig. 2) are not only the driest but also occur on the Edwards Plateau of central Texas, with distinct limestone soils. If edaphic variation rather than precipitation drives the symbiont $\times$ precipitation interaction observed in the field, we would expect the interaction to persist in the greenhouse, where soils from contrasting sites are brought into a common climate. Conversely, if the field pattern reflects a genuine climate-mediated effect of the symbiosis, the interaction should disappear when plants are decoupled from the climate of their soil's origin. To distinguish between these alternatives, we conducted a greenhouse experiment using *A. hyemalis* and soils collected from each field site, crossing symbiont status (S+/S-) with soil origin (seven sites; Fig. 2d; Table S1; Supplementary Methods S.1.2). Using data on biomass and inflorescence production in the greenhouse, we fitted similar hierarchical Bayesian models as the field analysis, with symbiont status, soil-origin precipitation (30-year normal, standardised), and their interaction as fixed effects (Supplementary Methods S.1.2.3).

Results

Efficacy of aridity and herbivory treatments

During the study period, realized precipitation at seven experimental sites corresponded well to 30-year climate normals, with wetter eastern sites and drier western sites (Fig. 2d). Virtually all host vital rates for all species declined with increasing aridity (see below), indicating that our geographically-distributed experiment effectively captured an abiotic stress gradient. Additionally, herbivore access/exclusion effectively modified biotic stress: plants in herbivore-access plots experienced 24% tiller damage on average, compared to 14% in exclusion plots (Fig. 2F, Fig. S2). S+ hosts experienced less herbivore damage than S- hosts in *A. hyemalis* (19% vs. 15%) and *P. autumnalis* (13% vs. 18%), consistent with hypothesized anti-herbivore benefits of endophyte symbiosis, but not in *E. virginicus* (44% vs. 42%; Fig. S4).

Endophyte effects on host fitness components across abiotic and biotic stress gradients

Annual survival was greater overall and more positively responsive to precipitation for the longer-lived hosts *E. virginicus* and *P. autumnalis* compared to the shorter-lived *A. hyemalis* (Fig. 3). Endophyte effects on survival depended most strongly on precipitation in *E. virginicus*. Under herbivore exclusion, effects of endophytes on survival shifted from strongly negative under dry conditions ($\Delta = -0.13$, $\Pr(\Delta > 0) = 0.02$) to positive under wetter conditions ($\Delta = 0.12$, $\Pr(\Delta > 0) = 0.97$). Under herbivore access the same directional shift occurred but with weaker magnitude ($\Delta = -0.06$ to 0.10, $\Pr(\Delta > 0) = 0.10$–0.89). A similar but weaker directional shift occurred in *A. hyemalis* under both herbivory treatments. For the host *P. autumnalis*, survival was strongly responsive to climate variation but there was no evidence that symbionts modified the response to climate stress, and there was little difference between herbivory treatments (Fig. 5, Fig. 3).

For growth, endophyte benefits increased with precipitation in *A. hyemalis* and *E. virginicus* under herbivore access, with strong positive effects of symbiosis under the wettest conditions (*A. hyemalis*: $\Delta = 0.49$, $\Pr(\Delta > 0) = 0.99$; *E. virginicus*: $\Delta = 0.32$, $\Pr(\Delta > 0) = 0.97$; Fig. 4). Under herbivore exclusion, *A. hyemalis* showed a similar positive trend but with weaker support ($\Delta = 0.29$, $\Pr(\Delta > 0) = 0.90$), while *E. virginicus* showed little variation across the gradient ($\Delta =$ 0.09–0.10, $\Pr(\Delta > 0) = 0.60$–0.75). As for survival, *P. autumnalis* growth rates declined strongly under drier conditions but endophyte effects were weak and highly uncertain (Fig. 5). Under herbivore exclusion, endophyte effects on *P. autumnalis* growth shifted from weakly positive to negative with increasing precipitation ($\Delta = -0.22$, $\Pr(\Delta > 0) = 0.13$), the only vital rate response that was even directionally consistent with drought-protective benefits of symbiosis.

For inflorescence production, endophyte effects were strongly positive in *P. autumnalis* and increased with precipitation under both herbivory treatments (Fig. S8). Under herbivore access, effects ranged from weak under drier conditions ($\Delta = 0.02$, $\Pr(\Delta > 0) = 0.68$) to strongly positive under wetter conditions ($\Delta = 2.91$, $\Pr(\Delta > 0) = 0.99$). Under herbivore exclusion, benefits were consistently strong across the gradient ($\Delta = 0.19$–4.97, $\Pr(\Delta > 0) = 0.97$–0.99). In contrast, endophyte effects on inflorescence production were weaker and highly uncertain across the gradient in the hosts *A. hyemalis* and *E. virginicus* under both herbivory treatments (Fig. S8; Fig. 5). Similarly, spikelet production in *E. virginicus* and *P. autumnalis* was strongly climate-sensitive, but there were no strong endophyte effects on spikelet production regardless of abiotic or biotic contexts.

Figure 5 synthesizes statistical confidence in endophyte effects across host species, vital rates, and abiotic and biotic stress gradients. The broad trend that emerged was a shift from negative or neutral endophyte effects under drier conditions to more positive effects under wetter conditions. These patterns were largely consistent with the hypothesis that abiotic (climate) stress is the dominant modifier of endophyte effects (Hypothesis 3; Fig. 1c), but in the opposite direction than predicted, as endophyte benefits tended to increase rather than decrease with precipitation. Partial support emerged for herbivory and combined effects of herbivory and climate as symbiosis modifiers (Hypotheses 2 and 4; Fig. 1b,d), depending on species and fitness component. Herbivory played a secondary role for most species and vital rates. Herbivore exclusion generally dampened the growth benefits that hosts gained from endophytes, consistent with an anti-herbivore protective role of *Epichloë* endophytes. However, the role of herbivory as a modifier of endophyte effects on other species and vital rates were more idiosyncratic.

Greenhouse experiment: soil legacy effects do not mediate endophyte responses

In the greenhouse experiment with *A. hyemalis* growing on soils from each experimental site, fitness components responded strongly to soil-origin precipitation even under uniform conditions: plants grown in soils from wetter sites produced greater aboveground biomass and more inflorescences than those from drier-origin soils (Figs. 6a–d), confirming that soil properties co-vary with the field aridity gradient. However, the symbiont $\times$ soil-origin precipitation interaction was negligible for both aboveground biomass (median $= 0.01$, 95% CI: -0.39–0.43, $P(>0) = 0.52$) and inflorescence count (median $= 0.19$, 95% CI: -0.39–0.72, $P(>0) = 0.76$; Fig. 6e). The absence of this interaction when plants are decoupled from their soil's source climate suggests that the symbiont $\times$ precipitation pattern observed in the field reflects a climate-mediated effect rather than a consequence of co-varying edaphic conditions.

Discussion

Drawing on three years of field experiments and Bayesian analysis of host demography, we evaluated how biotic and abiotic stressors jointly influence the demographic effects of vertically transmitted *Epichloë* endophytes on three cool-season perennial grasses across a natural climatic gradient. We predicted that endophytes would enhance host fitness under aridity and/or herbivore pressure (Fig. 1), consistent with stress amelioration and defense scenarios. Instead, endophyte-mediated benefits were strongest under benign, mesic conditions and weakened as abiotic stress intensified, while herbivory effects were inconsistent across species and fitness components, though the strongest herbivore effects were in the predicted direction (herbivore exclusion dampened endophyte benefits). These results contradict predictions of the Stress Gradient Hypothesis and instead support a contrasting general principle: the demographic benefits of symbionts are constrained by host condition, declining as abiotic stress reduces the resources available to maintain the association.

Endophyte presence is often assumed to enhance host performance under unfavorable environmental conditions, with meta-analyses reporting increased growth, reproduction, and survival across grass–*Epichloë* systems (Decunta et al., 2021). However, our results show that these benefits decline with increasing aridity. The mechanisms underlying this pattern remain uncertain, but several non-exclusive processes may contribute. Reduced water availability may limit the host's capacity to benefit from the symbiosis through constrained nutrient uptake, reduced carbon gain, or drought-induced hormonal shifts that impair endophyte function or alter the cost–benefit balance of the interaction (Ahlholm et al., 2002; Bastías et al., 2024; Bronstein, 2001; Cui et al., 2022; Faeth, 2002). This interpretation is consistent with the microbial mitigation–exacerbation continuum (David et al., 2018), which predicts that symbionts shift from beneficial to costly as environmental stress exceeds the host's capacity to sustain the interaction, and with empirical patterns along precipitation gradients in other grass–*Epichloë* systems (Kannadan and Rudgers, 2008). Because vertically transmitted endophytes cannot be expelled when conditions deteriorate, symbiotic hosts may be unable to avoid elevated costs under stress, in contrast to more flexible mutualisms (Afkhami and Rudgers, 2009).

Host identity strongly modulated the climate-dependence of endophyte benefits, with species showing divergent responses across the precipitation gradient. *P. autumnalis* maintained strong reproductive benefits regardless of climate, indicating relative buffering against abiotic stress, but weaker endophyte effects on survival and growth. In contrast, *A. hyemalis* and *E. virginicus* showed pronounced climate-dependent survival and growth responses, consistent with greater sensitivity to water limitation near their arid range edges. These contrasts may reflect differences in host physiological tolerance, symbiont identity, or the ecological contexts in which each species evolved (Decunta et al., 2021), and highlight the importance of host identity in determining the net outcome of grass–endophyte symbioses under climate stress (Cheplick and Faeth, 2009). Future studies could quantify these dynamics by measuring nonstructural carbohydrates in different tissues and across precipitation gradients, providing a clearer picture of how carbon limitation shapes symbiotic outcomes across host species and aridity gradients.

Herbivore exclusion consistently dampened endophyte benefits for growth across species, consistent with an anti-herbivore protective role of *Epichloë* endophytes that is reinforced under natural herbivore pressure (Clay, 1988; Decunta et al., 2021). Effects of herbivory on survival and reproduction were more idiosyncratic, varying across species and precipitation contexts, suggesting that the consequences of hosting a symbiont are not uniformly mediated by herbivore pressure across fitness components. The herbivore exclusion treatment confirmed that fencing effectively reduced herbivory across all three host species (Fig. S4), validating that these contrasts reflect genuine biological responses rather than treatment failure. Endophytes reduced herbivory damage in *A. hyemalis* and *P. autumnalis*, consistent with alkaloid-mediated defense, but increased damage in *E. virginicus* (Fig. S4), possibly because endophyte-symbiotic plants were more vigorous and therefore more attractive to herbivores. This species-specific reversal suggests that endophyte effects on herbivory depend on host condition rather than being a fixed symbiont trait, and that the demographic consequences of endophyte-mediated defense are filtered through the specific environmental currency limiting each fitness component.

The greenhouse experiment suggests that the endophyte–climate interaction observed in the field is more strongly linked to the direct physiological effects of water availability on the host than to soil properties that co-vary with aridity. When plants were grown in soils originating from contrasting sites under a common watering regime, the symbiont $\times$ soil-origin precipitation interaction was negligible for both aboveground biomass and inflorescence production (Van der Putten et al., 2013). These results suggest that differences in soil microbial composition, nutrient availability, or water retention are unlikely to be the primary drivers of the field patterns, although edaphic effects cannot be fully excluded given the inferential nature of the experiment. This decoupling instead supports the interpretation that endophyte costs and benefits depend primarily on contemporaneous water availability rather than on historical soil conditions associated with site aridity. Notably, plants originating from the driest sites performed poorly even under favorable watering conditions, suggesting that climatic and edaphic stressors may jointly constrain performance at range margins, while the endophyte component of this response is more directly linked to host physiological responses to water availability.

Our findings suggest that the degree of host control over symbiont retention is an underappreciated axis shaping mutualism outcomes along environmental gradients (Bronstein, 2001). Because *Epichloë* endophytes are tightly integrated with host reproduction and are not readily shed, hosts may continue to bear symbiont-associated costs even when environmental conditions reduce net benefits. Under such conditions, positive effects may emerge only when resource availability is sufficient to offset these costs. Our results therefore suggest that the stress-gradient hypothesis may require additional nuance when applied to symbioses in which hosts have limited flexibility to regulate symbiont load under stress (David et al., 2018). Historical and comparative evidence further suggests that environmental change can reshape symbiont prevalence and host–symbiont associations across diverse systems, including the breakdown of coral–algal symbioses during marine heat stress (Hoeksema and Bruna, 2000) and shifts in endophyte prevalence across grass populations over recent climate change (Fowler et al., 2025; Li et al., 2026). Our results provide a potential demographic mechanism for these patterns: as aridity intensifies, vertically transmitted symbionts may shift from beneficial to costly precisely where hosts are most vulnerable. Integrating host control over symbiont persistence and host physiological condition into predictions of symbiosis outcomes under climate change may therefore improve forecasts of whether grass–endophyte associations buffer or amplify demographic declines at range margins.

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Figures

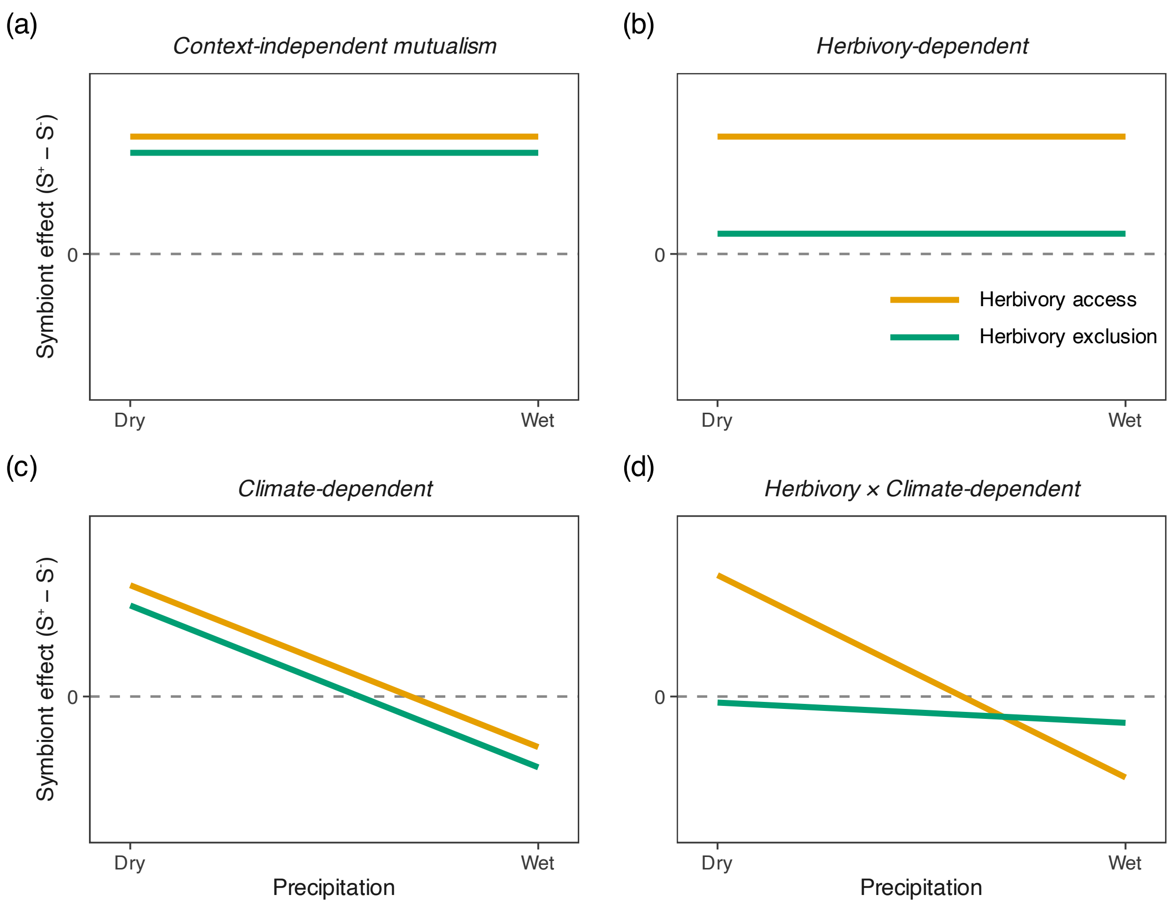

Figure 1: Competing hypotheses for how plant–symbiont interactions vary along a precipitation gradient under contrasting herbivory regimes. (a) Context-independent mutualism, where symbiont effects remain constant across environmental conditions and herbivory treatments. (b) Herbivory-dependent interactions, where the effect of symbionts differs in magnitude between herbivory access and exclusion but is insensitive to climate. (c) Climate-dependent interactions, where symbiont effects shift systematically along the precipitation gradient, independent of herbivory. (d) Combined herbivory- and climate-dependent interactions, where both environmental context and biotic pressure jointly shape symbiont effects, resulting in non-parallel and potentially decoupled response trajectories. Together, these scenarios illustrate alternative mechanisms by which symbiotic interaction outcomes may vary across abiotic and biotic stress gradients.

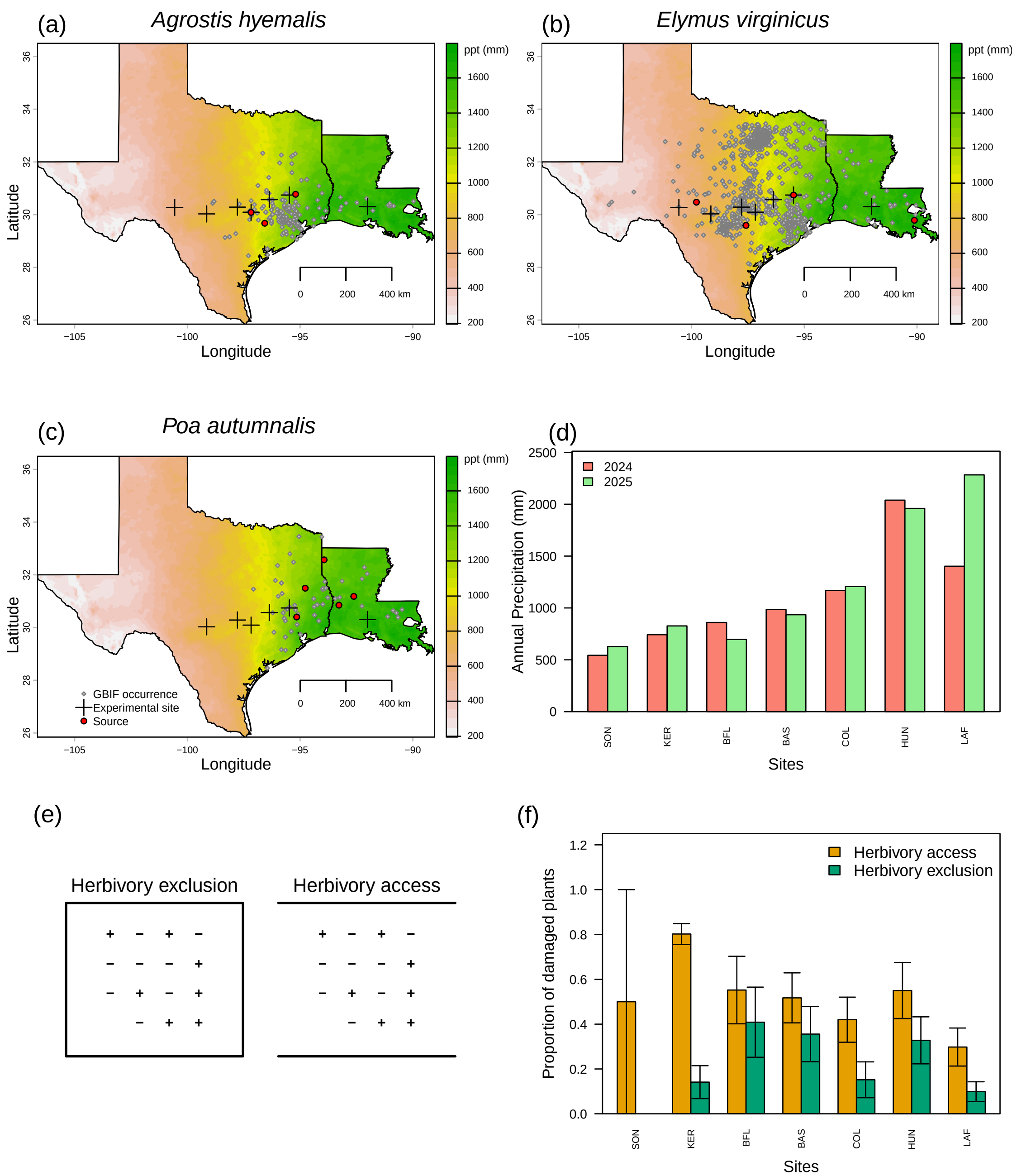

Figure 2: Geography of common garden sites along a natural aridity gradient from Texas to Louisiana, USA for (A) *Agrostis hyemalis*, (B) *Elymus virginicus*, and (C) *Poa autumnalis*. Climate data in A–C are based on 30-year normals for average annual precipitation (1996–2025) from PRISM (Hart and Bell, 2015), illustrating the precipitation gradient across experimental sites (scale bar in mm). Red dots indicate source population locations; grey dots represent GBIF occurrence records within the study area (GBIF, 2025a,b,c). (D) Annual cumulative precipitation during census years (2024 and 2025) at each experimental site, ordered west to east, derived from PRISM climate data. (E) Schematic illustration of herbivore exclusion and herbivore access plots; '+' denotes symbiont-positive and '-' denotes symbiont-negative individuals. Equal numbers are shown for illustration, although symbiont frequencies varied among experimental plots. (F) Proportion of damaged plants in herbivore exclusion and herbivore access plots across experimental sites (mean $\pm$ SE). Species-specific breakdowns are provided in Supplementary Figs S2 and S3.

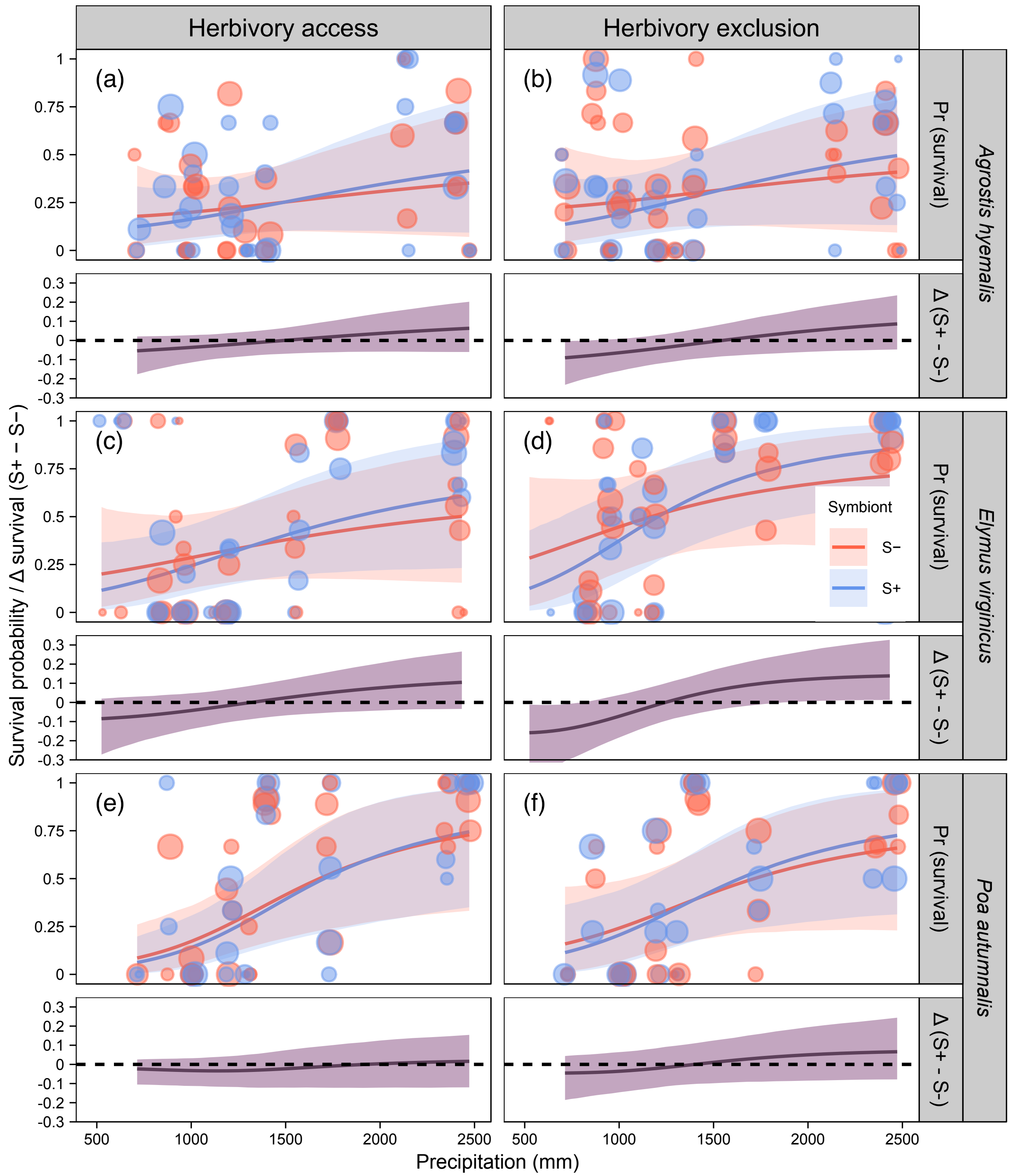

Figure 3: Predicted survival probability (upper panels) and symbiont effect (Δ = S+ − S-; lower panels) across the precipitation gradient. Lines and shaded regions show posterior means and 90% credible intervals, respectively. Points indicate observed plot means, with point size proportional to sample size (1–12 individuals per plot per symbiont status). Horizontal dashed line denotes $\Delta = 0$. Panels are arranged by species and herbivory treatment (herbivore access and exclusion).

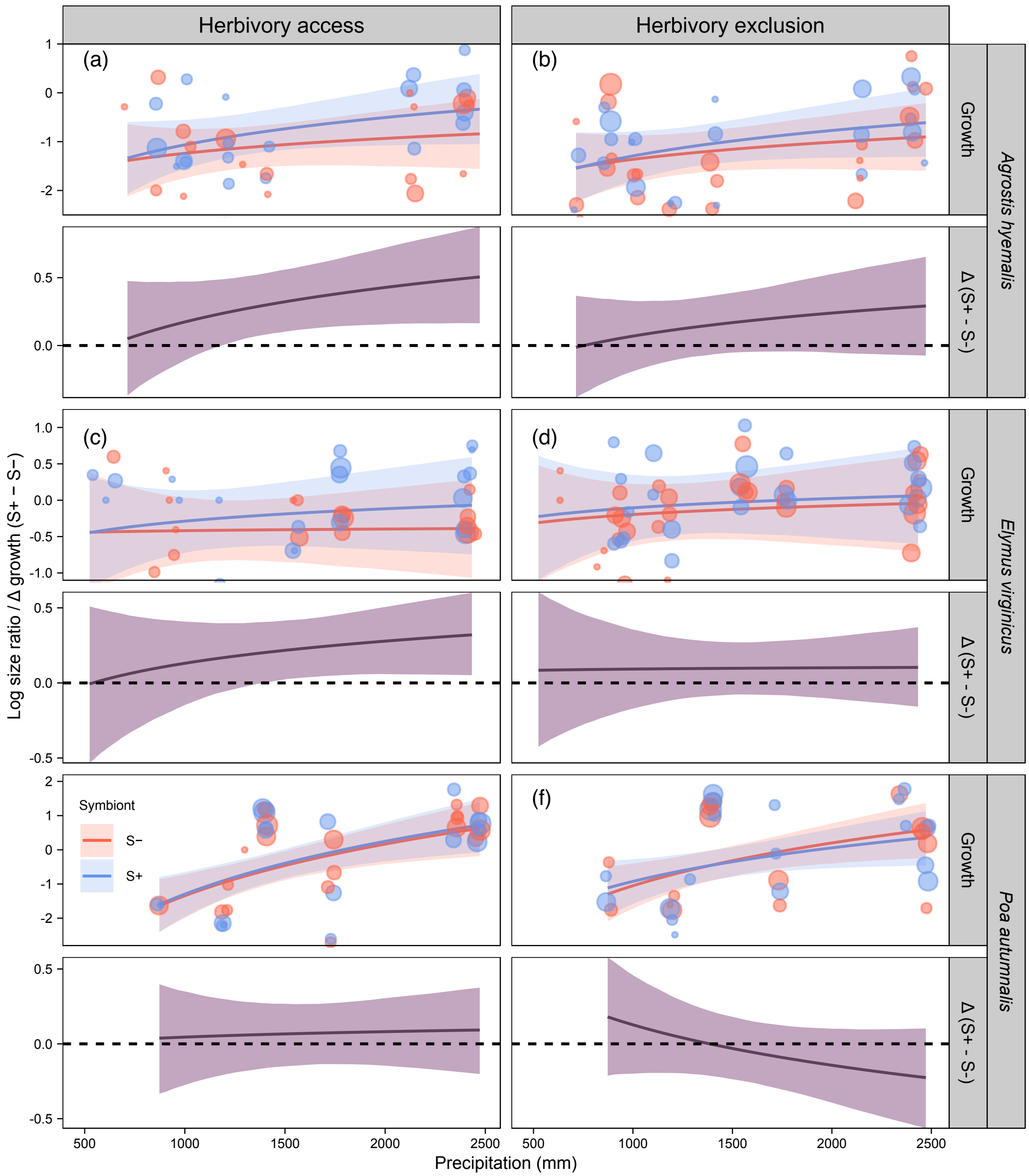

Figure 4: Predicted relative growth rate (upper panels) and symbiont effect (Δ = S+ − S-; lower panels) across the precipitation gradient. Lines and shaded regions show posterior means and 90% credible intervals, respectively. Points indicate observed plot means, with point size proportional to sample size (1–12 individuals per plot per symbiont status). Horizontal dashed line denotes $\Delta = 0$. Panels are arranged by species and herbivory treatment (herbivore access and exclusion).

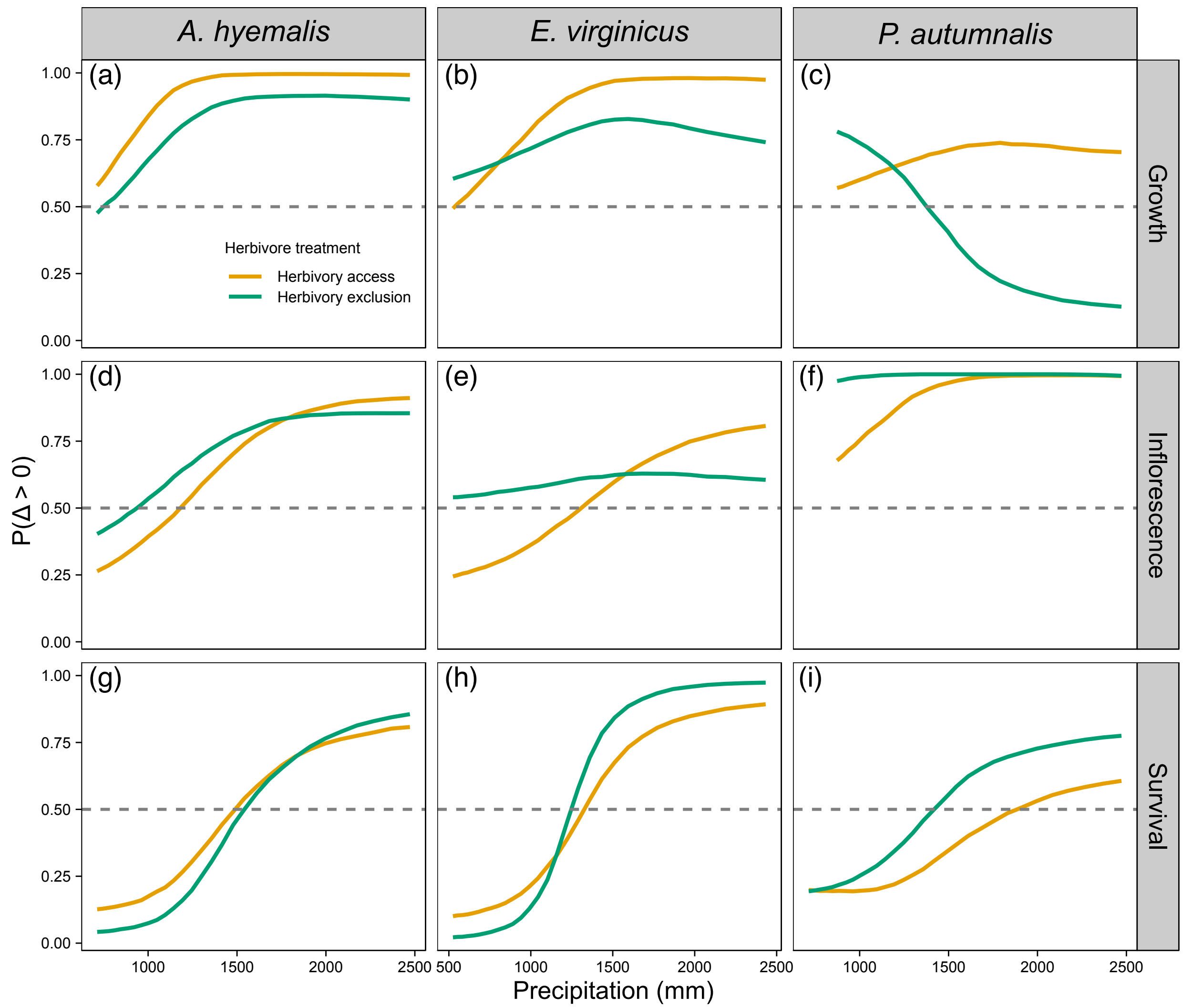

Figure 5: Probabilities of positive endophyte effects on survival, growth, and inflorescence production across precipitation gradients and under herbivore access or exclusion for *A. hyemalis*, *E. virginicus*, and *P. autumnalis*. Panels show the probability that the difference in vital rates between S+ and S- hosts is positive ($\Pr[\Delta > 0]$). Values closer to 1.0 indicate high confidence in an S+ advantage, values closer to 0.0 indicate high confidence in an S- advantage, and values close to 0.5 indicate no difference between S+ and S-.

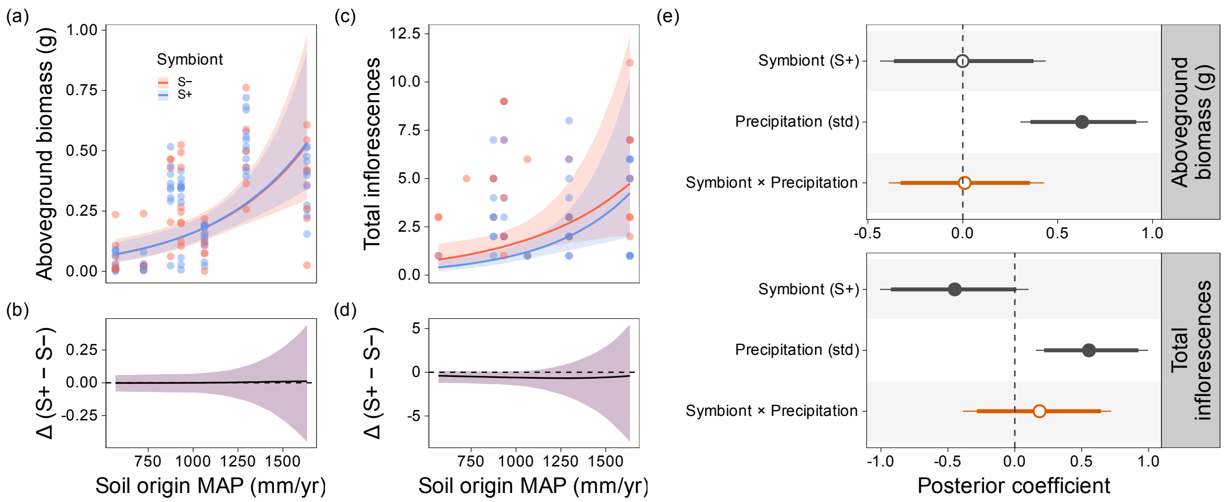

Figure 6: Greenhouse experiment testing whether soil-origin precipitation modifies symbiont effects on host performance in *Agrostis hyemalis*. Left and middle columns show posterior predictions for aboveground biomass and total inflorescence production across the 30-year mean annual precipitation of the soil-origin site (mm yr^{-1}) for symbiont-present (S+) and symbiont-absent (S-) plants. Points are observed values, lines are posterior medians, and shaded ribbons are 95% credible intervals. Lower panels show the posterior contrast between symbiont treatments (Δ = S+ $-$ S-). The right column shows posterior coefficient estimates (median $\pm$ 90% and 95% credible intervals) for symbiont presence, precipitation, and their interaction. Precipitation strongly increased performance, whereas the Symbiont $\times$ Precipitation interaction was weak.

S.1 Supporting Information

S.1.1 Supporting Methods

S.1.1.1 Endophyte status verification

To assess the efficacy of heat treatment and to confirm endophyte status before transfer to the field experiment, we verified symbiont status (S+ or S-) of plants from control and heat lineages using three methods. First, peels of the inner leaf sheaths were stained with aniline blue lactic acid and viewed under a compound microscope at 200x–400x to identify *Epichloë* hyphae (Jackson and Johnson-Cicalese, 1988). Second, we used microscopy to assess the presence or absence of *Epichloë* in seeds that were soaked in NaOH, then squashed and stained similarly to the leaf tissue, to determine the endophyte status of the maternal plant (Fowler et al., 2025). Third, we used an immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co., Watkinsville, Georgia, USA) that targets proteins of *Epichloë* spp. to visually indicate presence or absence (Koh et al., 2006). In total, we tested 1,481 plants for endophyte status. In cases where two or more methods were used on the same plants ($N = 253$), there was 99.2% agreement between methods. For genotypes that were vegetatively propagated, we tested the endophyte status of a single clonal replicate and assume that its status applied equally to all clones.

All S- plants used in the experiment came from heated-treated lineages, while S+ plants came from both control and heat lineages depending on the efficacy of the removal treatment. While heat treatment was effective at reducing endophyte prevalence relative to controls, our confidence that individual S- plants were experimentally disinfected, as opposed to naturally symbiont-free, varied due to among-population variation in endophyte prevalence (mean: 68% S+) and the effectiveness of the heat treatment (Table S2). For example, for the SJC population of *A. hyemalis*, control and heat lineages yielded plants that were 100% and 5% S+, respectively – giving us high confidence that S- plants are derived from successful symbiont removal. For the SFAEF population of *P. autumnalis*, on the other hand, control and heat lineages yielded plants that were 35% and 26% S+, respectively – meaning that these S- plants were quite likely from naturally S- lineages.

S.1.2 Greenhouse experiment

S.1.2.1 Experimental design

To examine whether edaphic factors mediate plant performance across the precipitation gradient, we conducted a greenhouse experiment manipulating soil origin and endophyte status.

Soils were collected during summer 2024 from the seven field sites used in the range-limits experiment. These sites span the precipitation gradient from the wettest site to the driest site. After collection, soils were stored in a cold room at 3.3°C until the start of the experiment. Soils were not sterilized and therefore soil origin treatment reflects both the physical properties of the soil as well as the microbial communities that may vary across soil types.

Seeds of *Agrostis hyemalis* were obtained from the seed stock used to generate plants in the range-limits field experiment. Seeds originated from two source populations (LPEF and SJC) and were grouped according to endophyte status as endophyte-symbiotic (S+) or endophyte-free (S-). Seeds were germinated on agar plates (18.2 g agar per 1000 mL deionized water) in a growth chamber and subsequently transplanted as seedlings into pots containing soils from each field site. Plants were randomly assigned to greenhouse trays and maintained under controlled greenhouse conditions with automated irrigation. The experimental design crossed seed source population, endophyte status, and soil origin. Realised sample sizes varied across treatment combinations due to mortality and poor germination and are reported in Table S5.

S.1.2.2 Data collection

Plants were monitored regularly for survival and reproductive development. Reproductive output was quantified as inflorescence number and spikelet production. During the flowering season, plants were checked at least twice per week for the presence of seed-bearing inflorescences. Mature inflorescences were collected as they developed to avoid repeated counts. For each flowering individual, spikelets were counted on at least three inflorescences when available. For individuals producing many inflorescences, mean spikelet number per inflorescence was used to estimate total spikelet production. After eight months of growth, plants were harvested to quantify aboveground biomass. Plant material was dried at 60°C in a drying oven and weighed using an analytical balance. Because inflorescences were collected prior to harvest, they were dried separately and added to the total aboveground biomass.

S.1.2.3 Statistical analysis

For each fitness component (aboveground biomass, inflorescence count, and spikelets per inflorescence), we fitted hierarchical Bayesian models mirroring the field analysis. Fixed effects included symbiont status (S+/S-), the 30-year mean annual precipitation of each soil's origin site (standardised; hereafter soil-origin MAP), and their interaction, with planting date (standardised) as a covariate. Source population was included as a random effect with a non-centred parameterisation. Biomass was log-transformed and modelled with a Student-t likelihood ($\nu = 3$) to provide robustness to outliers, which are common in small greenhouse experiments; count responses (inflorescence number and spikelets per inflorescence) used negative binomial likelihoods to accommodate overdispersion. For all models, fixed-effect coefficients received Normal(0,1) priors; the overdispersion parameter for count models was placed on the log scale with a Normal(2,1) prior (centred on $\phi \approx 7$); and the random-effect standard deviation received an Exponential(2) prior.

Models were fitted in Stan (Stan Development Team, 2024) with four chains of 2,000 iterations (1,000 warmup; adapt_delta $= 0.99$). Convergence was assessed via $\hat{R} < 1.01$ for all parameters and inspection of trace plots. We report posterior medians with 95% credible intervals for three focal parameters: the symbiont main effect (S+ vs. S- at mean soil-origin MAP), the soil-origin MAP main effect, and the symbiont $\times$ soil-origin MAP interaction . To visualise how the endophyte effect varies across the soil-origin precipitation gradient, we computed the posterior contrast (S+ $-$ S-) at each point along the observed precipitation range, marginalising over planting date (set to the mean) and source population random effects. These contrasts are summarised as posterior medians with 95% credible intervals and displayed alongside the predicted responses for each symbiont status. Posterior predictive checks indicated adequate fit for all models (Fig. S6).

Supporting Tables

Species	Population	Latitude	Longitude	Notes	Total plants	Unique genotypes	Mean clones per genotype
A. hyemalis	SJC	30.771969	-95.200620	Private property, San Jacinto County, TX	273	201	1.36
A. hyemalis	COHR	29.672833	-96.567517	Roadside collection, Columbus, TX	289	35	8.26
A. hyemalis	LPEF	30.085383	-97.172508	Stengl Lost Pines Field Station, Bastrop, TX	278	41	6.78
P. autumnalis	LA7	30.849660	-93.276772	Roadside population, DeRidder, LA	91	9	10.11
P. autumnalis	LA1	32.568125	-93.932976	Walter P. Jacobs Nature Center, Shreveport, LA	129	7	18.43
P. autumnalis	LA2	31.183040	-92.616969	Kistachie National Forest, LA	253	15	16.87
P. autumnalis	SFAEF	31.492158	-94.772200	Stephen F. Austin Experimental Forest, TX	247	41	6.02
E. virginicus	HUNT	30.741797	-95.474508	Pineywoods Environmental Research Laboratory, Huntsville, TX	91	61	1.49
E. virginicus	RL5	30.471717	-99.783803	Llano River Field Station, Junction, TX	131	91	1.44
E. virginicus	PALM	29.591623	-97.584781	Palmetto State Park, TX	442	195	2.27
E. virginicus	JLP	29.785105	-90.114933	Jean Laffitte Park, LA	176	102	1.73

Table S1: Geographic coordinates and descriptions of source populations for study species. For each source population, the table shows the total number of plants included in the experiment, the number of unique genotypes, and the average clonal replication per genotype.

Table S2: Summary of symbiont removal treatments by species and population. N indicates the number of plants tested for symbiont status and %S+ indicates the proportion that tested positive. Treatment refers to the symbiont removal treatment applied to the seed lineage: Control (no treatment) or Heat (5 days in a drying oven at 60°C for *E. virginicus* and *P. autumnalis*; 12 minutes in a 60°C water bath for *A. hyemalis*). For all populations except SFAEF and PALM, heat-treated seed lineages were one or more generations removed from the treatment; for those two populations, plants from heat-treated seeds were used directly in the experiment.

Species	Population	Treatment	N	%S+
A. hyemalis	COHR	Control	84	0.92
A. hyemalis	COHR	Heat	75	0.04
A. hyemalis	LPEF	Control	7	0.57
A. hyemalis	LPEF	Heat	88	0.00
A. hyemalis	SJC	Control	132	1.00
A. hyemalis	SJC	Heat	191	0.05
E. virginicus	HUNT	Control	71	0.61
E. virginicus	HUNT	Heat	38	0.21
E. virginicus	JLP	Control	27	1.00
E. virginicus	JLP	Heat	113	0.02
E. virginicus	PALM	Control	176	0.61
E. virginicus	PALM	Heat	238	0.71
E. virginicus	RL5	Control	68	0.43
E. virginicus	RL5	Heat	74	0.73
P. autumnalis	LA1	Control	5	0.20
P. autumnalis	LA1	Heat	4	0.00
P. autumnalis	LA2	Control	10	0.80
P. autumnalis	LA2	Heat	3	0.33
P. autumnalis	LA7	Control	5	1.00
P. autumnalis	LA7	Heat	11	0.00
P. autumnalis	SFAEF	Control	23	0.35
P. autumnalis	SFAEF	Heat	38	0.26

Table S3: Common garden sites used for reciprocal transplant and environmental manipulation studies.

Site	Code	Latitude	Longitude
University of Louisiana Ecology Center	LAF	30.304773	-92.012401
Center for Biological Field Studies	HUN	30.742815	-95.476881
TAMU Ecology and Natural Resource Teaching Area	COL	30.569302	-96.366324
UT Stengl Lost Pines Field Station	BAS	30.092343	-97.172442
Private property, Kerrville, TX	KER	30.028437	-99.142888
Brackenridge Field Laboratory	BFL	30.284866	-97.780675
TAMU Sonora Station	SON	30.273467	-100.561463

Table S4: Comparison of candidate models (quadratic vs. linear) for each vital rate based on leave-one-out cross-validation (LOO-CV). ΔELPD is relative to the best model.

Vital rate	Model type	ΔELPD	SE
Survival	Linear	0.000	0.000
Survival	Quadratic	-0.282	0.663
Growth	Linear	0.000	0.000
Growth	Quadratic	-0.078	0.430
Inflorescence	Linear	0.000	0.000
Inflorescence	Quadratic	-0.689	0.516
Spikelet	Linear	0.000	0.000
Spikelet	Quadratic	-0.332	0.579

Table S5: Sample sizes for each combination of seed source population, soil-origin site, and endophyte treatment in the greenhouse experiment. The experiment crossed two seed source populations (LPEF and SJC), seven soil-origin sites along the precipitation gradient, and two symbiont treatments (S- and S+). Most treatment combinations included seven replicates, except LPEF $\times$ BFL $\times$ S- where $n = 6$ due to early mortality.

Response	Site	S-	S+
Biomass			
	LAF	14	14
	SON	14	14
	KER	14	14
	BFL	13	14
	BAS	14	14
	COL	14	14
	HUN	14	14
Inflorescence			
	LAF	14	14
	SON	14	14
	KER	14	14
	BFL	14	14
	BAS	14	14
	COL	14	14
	HUN	14	14
Spikelets			
	LAF	7	10
	SON	3	1
	KER	1	0
	BFL	3	8
	BAS	7	6
	COL	2	2
	HUN	5	12

Table S6: Posterior summaries of Δ (S+ − S−) for survival across precipitation quantiles (mm, 5%, 25%, 50%, 75%, 100%) for each species and herbivore treatment. Positive median values indicate beneficial effects of symbionts; negative medians indicate costly effects. $P(\Delta > 0)$ quantifies the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect. Herbivore treatment: Exclusion = herbivores excluded; Access = herbivores had access.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	P(Δ > 0)
A. hyemalis	Access	714.00	-0.04	-0.18	0.02	0.13
A. hyemalis	Access	963.00	-0.03	-0.12	0.03	0.16
A. hyemalis	Access	1300.00	-0.01	-0.08	0.05	0.35
A. hyemalis	Access	1832.00	0.02	-0.06	0.12	0.70
A. hyemalis	Access	2472.00	0.06	-0.06	0.20	0.81
A. hyemalis	Exclusion	714.00	-0.07	-0.23	-0.00	0.04
A. hyemalis	Exclusion	963.00	-0.06	-0.17	0.01	0.07
A. hyemalis	Exclusion	1300.00	-0.03	-0.11	0.04	0.25
A. hyemalis	Exclusion	1832.00	0.03	-0.07	0.14	0.70
A. hyemalis	Exclusion	2472.00	0.08	-0.05	0.24	0.86
E. virginicus	Access	525.00	-0.06	-0.27	0.02	0.10
E. virginicus	Access	760.00	-0.06	-0.20	0.03	0.13
E. virginicus	Access	1101.00	-0.03	-0.12	0.06	0.29
E. virginicus	Access	1680.00	0.04	-0.06	0.15	0.77
E. virginicus	Access	2433.00	0.10	-0.03	0.27	0.89
E. virginicus	Exclusion	525.00	-0.13	-0.40	-0.01	0.02
E. virginicus	Exclusion	760.00	-0.12	-0.29	-0.01	0.04
E. virginicus	Exclusion	1101.00	-0.04	-0.14	0.06	0.24
E. virginicus	Exclusion	1680.00	0.09	-0.02	0.21	0.91
E. virginicus	Exclusion	2433.00	0.12	0.01	0.33	0.97
P. autumnalis	Access	714.00	-0.01	-0.10	0.03	0.20
P. autumnalis	Access	963.00	-0.03	-0.11	0.03	0.19
P. autumnalis	Access	1300.00	-0.03	-0.12	0.05	0.26
P. autumnalis	Access	1832.00	-0.00	-0.12	0.11	0.49
P. autumnalis	Access	2472.00	0.01	-0.12	0.15	0.61
P. autumnalis	Exclusion	714.00	-0.03	-0.19	0.04	0.19
P. autumnalis	Exclusion	963.00	-0.03	-0.15	0.06	0.24
P. autumnalis	Exclusion	1300.00	-0.01	-0.11	0.10	0.41
P. autumnalis	Exclusion	1832.00	0.04	-0.09	0.17	0.70
P. autumnalis	Exclusion	2472.00	0.06	-0.08	0.24	0.78

Table S7: Posterior summaries of Δ (S+ − S-) for growth across precipitation quantiles (mm, 5%, 25%, 50%, 75%, 100%)) for each species and herbivore treatment. Positive median values indicate beneficial effects of symbionts; negative medians indicate costly effects. $P(\Delta > 0)$ quantifies the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	P(Δ > 0)
A. hyemalis	Access	714.00	0.05	-0.37	0.47	0.58
A. hyemalis	Access	963.00	0.15	-0.14	0.47	0.81
A. hyemalis	Access	1300.00	0.27	0.05	0.51	0.98
A. hyemalis	Access	1832.00	0.39	0.15	0.66	1.00
A. hyemalis	Access	2472.00	0.49	0.16	0.88	0.99
A. hyemalis	Exclusion	714.00	-0.01	-0.38	0.37	0.48
A. hyemalis	Exclusion	963.00	0.06	-0.21	0.33	0.65
A. hyemalis	Exclusion	1300.00	0.14	-0.08	0.35	0.85
A. hyemalis	Exclusion	1832.00	0.22	-0.04	0.48	0.91
A. hyemalis	Exclusion	2472.00	0.29	-0.07	0.65	0.90
E. virginicus	Access	525.00	-0.00	-0.53	0.51	0.49
E. virginicus	Access	760.00	0.07	-0.31	0.44	0.64
E. virginicus	Access	1101.00	0.15	-0.10	0.40	0.85
E. virginicus	Access	1680.00	0.24	0.04	0.44	0.98
E. virginicus	Access	2433.00	0.32	0.05	0.60	0.97
E. virginicus	Exclusion	525.00	0.09	-0.42	0.61	0.60
E. virginicus	Exclusion	760.00	0.09	-0.27	0.45	0.65
E. virginicus	Exclusion	1101.00	0.09	-0.14	0.33	0.75
E. virginicus	Exclusion	1680.00	0.10	-0.08	0.28	0.82
E. virginicus	Exclusion	2433.00	0.10	-0.16	0.37	0.74
P. autumnalis	Access	873.00	0.04	-0.33	0.40	0.57
P. autumnalis	Access	1122.00	0.06	-0.23	0.31	0.63
P. autumnalis	Access	1443.00	0.07	-0.16	0.27	0.70
P. autumnalis	Access	1923.00	0.08	-0.14	0.29	0.73
P. autumnalis	Access	2472.00	0.10	-0.20	0.38	0.70
P. autumnalis	Exclusion	873.00	0.17	-0.21	0.58	0.78
P. autumnalis	Exclusion	1122.00	0.08	-0.19	0.37	0.68
P. autumnalis	Exclusion	1443.00	-0.02	-0.22	0.20	0.45
P. autumnalis	Exclusion	1923.00	-0.13	-0.36	0.11	0.19
P. autumnalis	Exclusion	2472.00	-0.22	-0.56	0.10	0.13

Table S8: Posterior summaries of Δ (S+ − S-) for inflorescence production across precipitation quantiles (mm, 5%, 25%, 50%, 75%, 100%) for each species and herbivore treatment. Positive median values indicate beneficial effects of symbionts; negative medians indicate costly effects. $P(\Delta > 0)$ quantifies the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	P(Δ > 0)
A. hyemalis	Access	714.00	-0.08	-0.61	0.18	0.26
A. hyemalis	Access	963.00	-0.06	-0.54	0.30	0.37
A. hyemalis	Access	1300.00	0.06	-0.44	0.57	0.59
A. hyemalis	Access	1832.00	0.46	-0.31	1.85	0.85
A. hyemalis	Access	2472.00	1.30	-0.29	6.06	0.91
A. hyemalis	Exclusion	714.00	-0.06	-0.74	0.54	0.40
A. hyemalis	Exclusion	963.00	0.01	-0.63	0.69	0.51
A. hyemalis	Exclusion	1300.00	0.19	-0.50	1.07	0.70
A. hyemalis	Exclusion	1832.00	0.67	-0.49	2.62	0.84
A. hyemalis	Exclusion	2472.00	1.48	-0.95	7.71	0.85
E. virginicus	Access	525.00	-0.09	-0.67	0.16	0.24
E. virginicus	Access	760.00	-0.08	-0.48	0.18	0.29
E. virginicus	Access	1101.00	-0.04	-0.35	0.23	0.41
E. virginicus	Access	1680.00	0.07	-0.24	0.48	0.67
E. virginicus	Access	2433.00	0.25	-0.27	1.36	0.81
E. virginicus	Exclusion	525.00	0.03	-0.87	1.17	0.54
E. virginicus	Exclusion	760.00	0.04	-0.68	0.92	0.56
E. virginicus	Exclusion	1101.00	0.07	-0.52	0.74	0.59
E. virginicus	Exclusion	1680.00	0.09	-0.45	0.75	0.63
E. virginicus	Exclusion	2433.00	0.11	-0.97	1.40	0.60
P. autumnalis	Access	873.00	0.02	-0.07	0.20	0.68
P. autumnalis	Access	1122.00	0.07	-0.06	0.38	0.82
P. autumnalis	Access	1443.00	0.26	0.01	0.83	0.96
P. autumnalis	Access	1923.00	0.98	0.24	2.89	0.99
P. autumnalis	Access	2472.00	2.91	0.48	11.89	0.99
P. autumnalis	Exclusion	873.00	0.19	0.02	1.00	0.97
P. autumnalis	Exclusion	1122.00	0.44	0.10	1.42	1.00
P. autumnalis	Exclusion	1443.00	0.97	0.35	2.44	1.00
P. autumnalis	Exclusion	1923.00	2.37	0.75	6.35	1.00
P. autumnalis	Exclusion	2472.00	4.97	0.91	19.92	0.99

Table S9: Posterior summaries of Δ (S+ − S-) for spikelet production across precipitation quantiles (mm, 5%, 25%, 50%, 75%, 100%) for each species and herbivore treatment. Positive median values indicate beneficial effects of symbionts; negative medians indicate costly effects. $P(\Delta > 0)$ quantifies the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	P(Δ > 0)
A. hyemalis	Access	525.00	-2.33	-9.38	3.12	0.23
A. hyemalis	Access	760.00	-1.84	-7.08	2.69	0.24
A. hyemalis	Access	1101.00	-1.19	-4.99	2.31	0.28
A. hyemalis	Access	1680.00	-0.27	-3.26	2.81	0.44
A. hyemalis	Access	2433.00	0.74	-3.23	4.96	0.62
A. hyemalis	Exclusion	525.00	-5.51	-14.68	1.98	0.11
A. hyemalis	Exclusion	760.00	-3.70	-9.87	1.93	0.14
A. hyemalis	Exclusion	1101.00	-1.69	-5.49	2.03	0.23
A. hyemalis	Exclusion	1680.00	0.83	-1.56	3.36	0.71
A. hyemalis	Exclusion	2433.00	3.20	-0.17	7.60	0.94
E. virginicus	Access	873.00	-0.76	-5.08	3.35	0.35
E. virginicus	Access	1122.00	-0.52	-3.80	2.75	0.38
E. virginicus	Access	1443.00	-0.18	-2.60	2.28	0.45
E. virginicus	Access	1923.00	0.31	-1.88	2.57	0.59
E. virginicus	Access	2472.00	0.84	-2.35	4.34	0.67
E. virginicus	Exclusion	873.00	2.73	-2.78	9.74	0.81
E. virginicus	Exclusion	1122.00	2.96	-1.31	8.23	0.88
E. virginicus	Exclusion	1443.00	3.13	0.02	6.92	0.95
E. virginicus	Exclusion	1923.00	3.35	0.62	6.84	0.98
E. virginicus	Exclusion	2472.00	3.46	-0.60	8.59	0.92

Model	Parameter	Median	Lower 95%	Upper 95%	P(>0)
Biomass (Gaussian)	Symbiont status (S+)	-0.001	-0.436	0.436	0.498
Biomass (Gaussian)	Symbiont status (S+) × Precipitation	0.010	-0.390	0.428	0.517
Inflo count (NB)	Symbiont status (S+)	-0.449	-1.007	0.104	0.054
Inflo count (NB)	Symbiont status (S+) × Precipitation	0.185	-0.388	0.719	0.757
Total spikelets projected (NB)	Symbiont status (S+)	-0.158	-0.550	0.233	0.208
Total spikelets projected (NB)	Symbiont status (S+) × Precipitation	-0.130	-0.459	0.184	0.214

Table S10: Posterior summaries of the interaction between endophyte presence (S+) and source population (Site) for each fitness component. For each model, the table reports the median estimate, 95% credible interval (Lower and Upper), and the posterior probability that the effect is greater than zero (P(>0)).

667 ### S.1.3 Supporting Figures

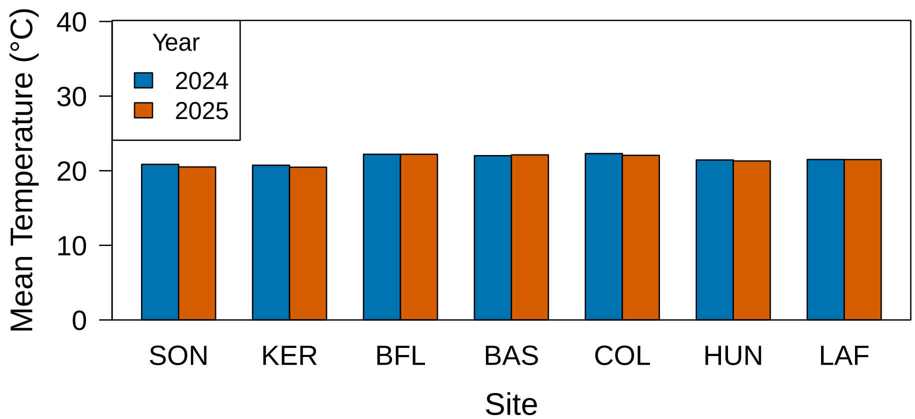

Fig S1: Annual temperature (°C) during the 2024 and 2025 census years. Sites are arranged west to east by longitude.

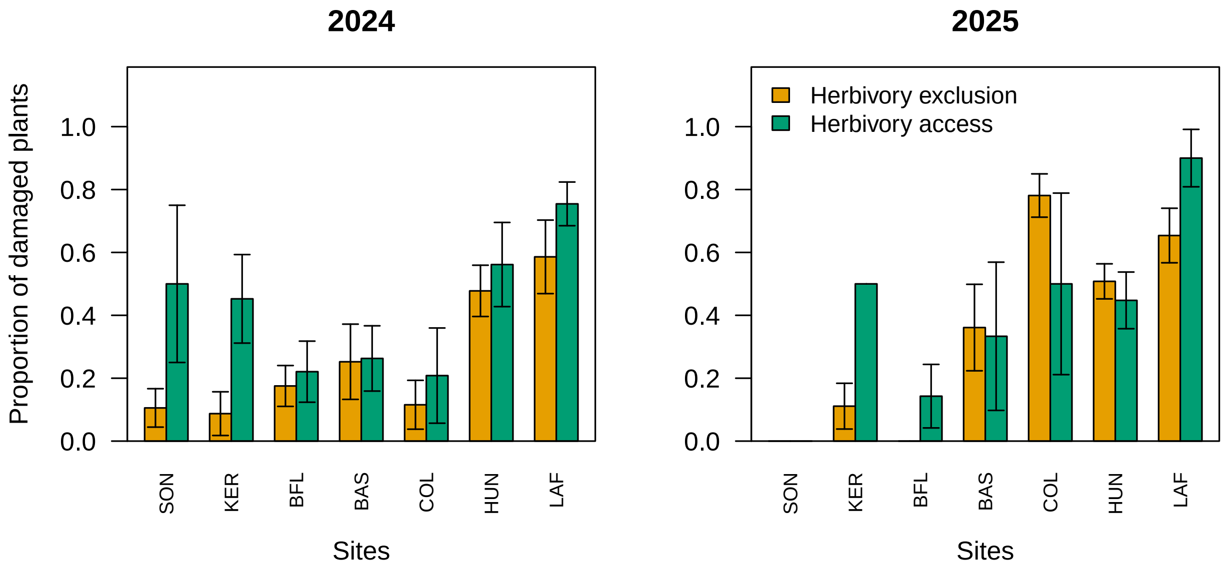

Fig S2: Proportion of plants damaged by herbivory across sites in 2024 and 2025. Barplots show fenced (dark) versus unfenced (grey) plots for each site (ordered west to east). Error bars indicate standard errors across plots within each site. Panels show results for 2024 (left) and 2025 (right).

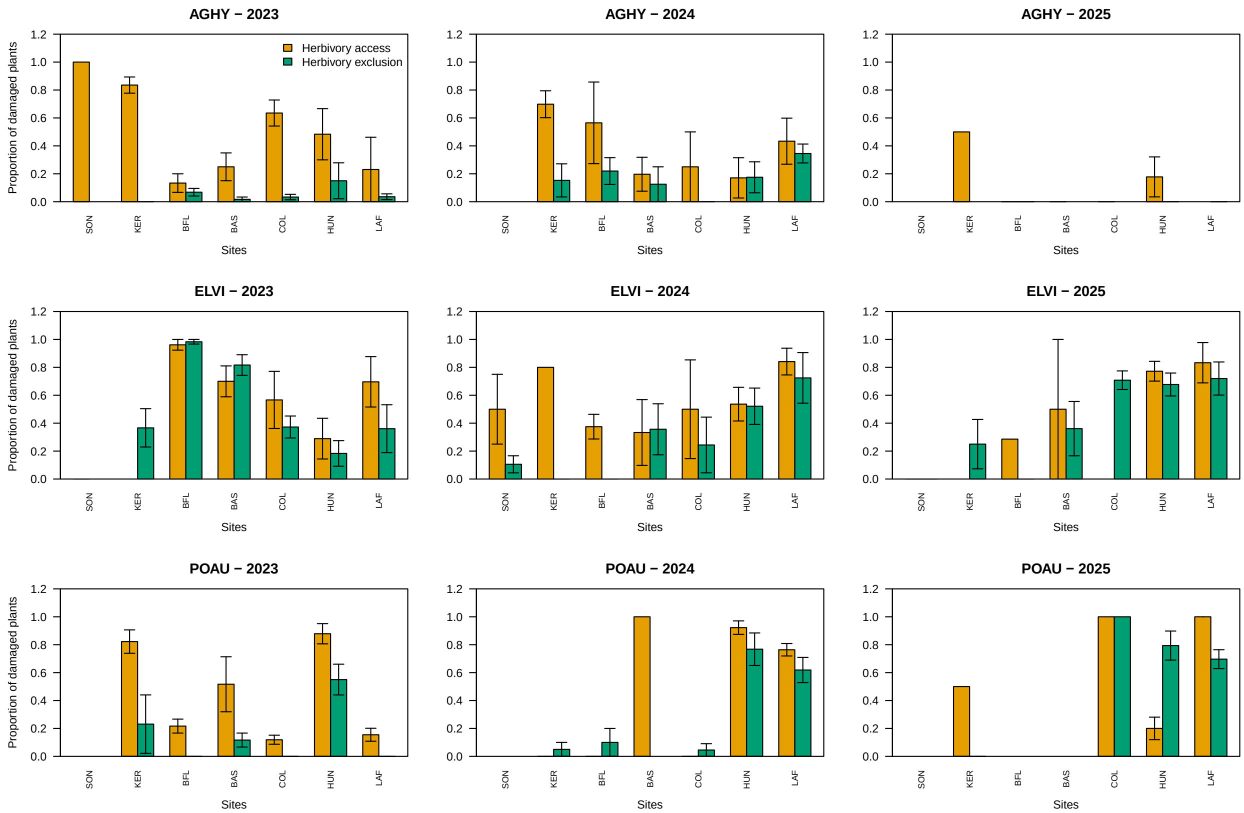

Fig S3: Proportion of plants damaged by herbivory across sites for *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* over 2023–2025. Barplots show fenced (grey80) versus unfenced (grey15) proportions for each site (ordered west to east: SON, KER, BFL, BAS, COL, HUN, LAF). Error bars indicate standard errors across plots within each site. Panels are arranged by species (rows) and year (columns).

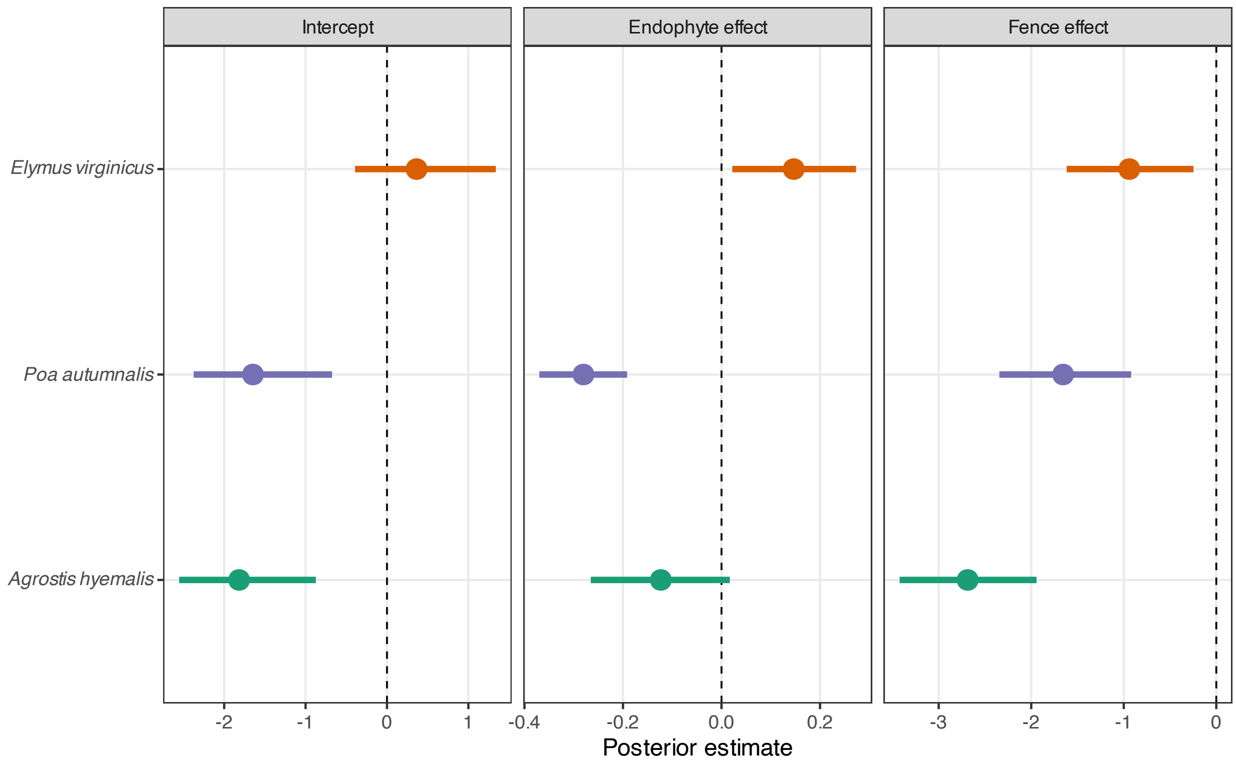

Fig S4: Posterior estimates (median and 90% credible intervals) of fence and endophyte effects on the proportion of damaged tillers for three host grass species, from a binomial model with site, year, and plot as random effects. Negative values indicate reduced herbivory damage; positive values indicate increased damage. Points and intervals are colored by species: *Agrostis hyemalis* (green), *Elymus virginicus* (orange), and *Poa autumnalis* (purple). Dashed vertical lines indicate no effect.

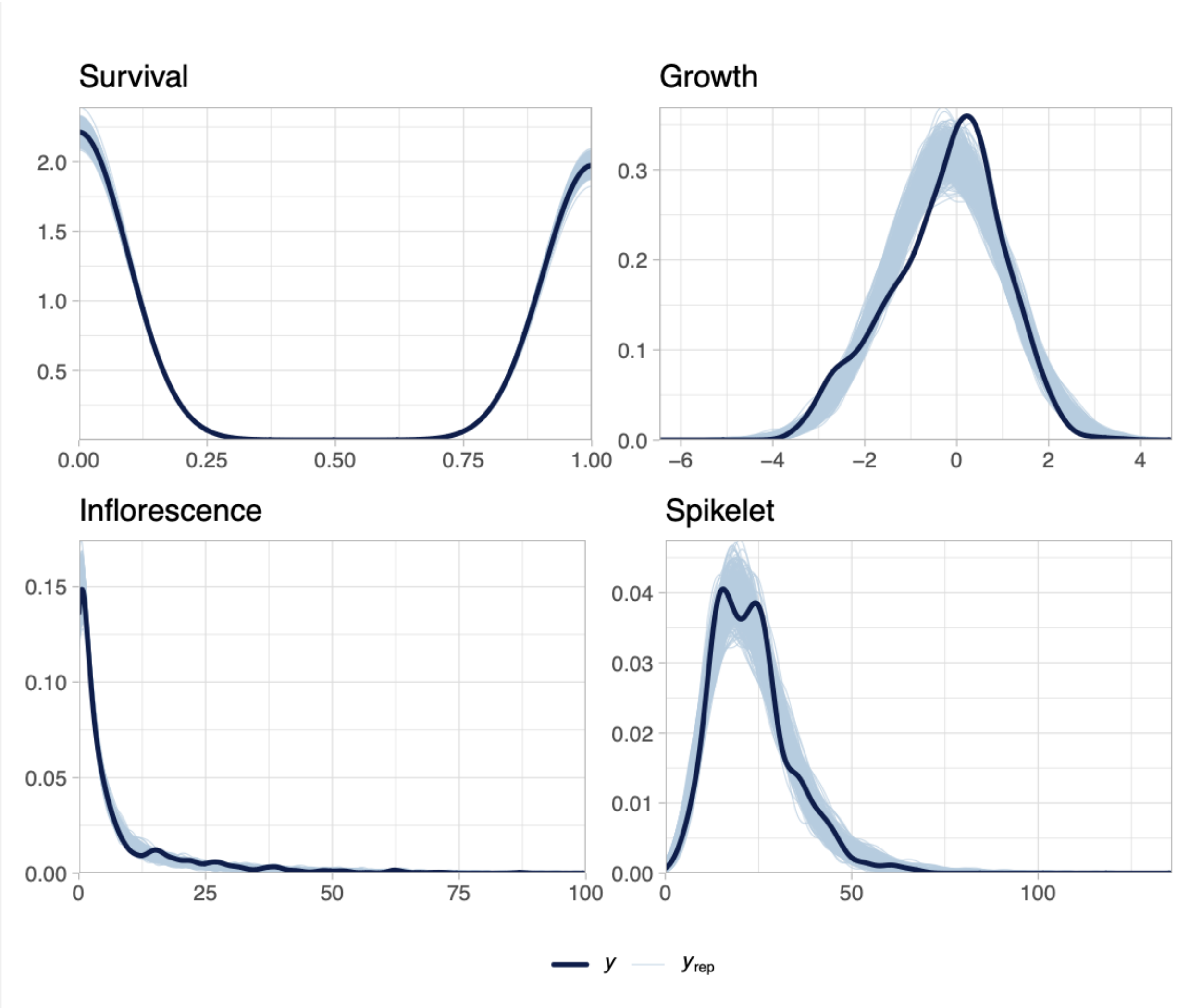

Fig S5: Posterior predictive checks for each vital rate. Dark blue lines show observed data (y) and light blue lines show replicated datasets from posterior predictions (y_{rep}). Strong overlap across survival, growth, inflorescence, and spikelet production indicates adequate model fit for all vital rates.

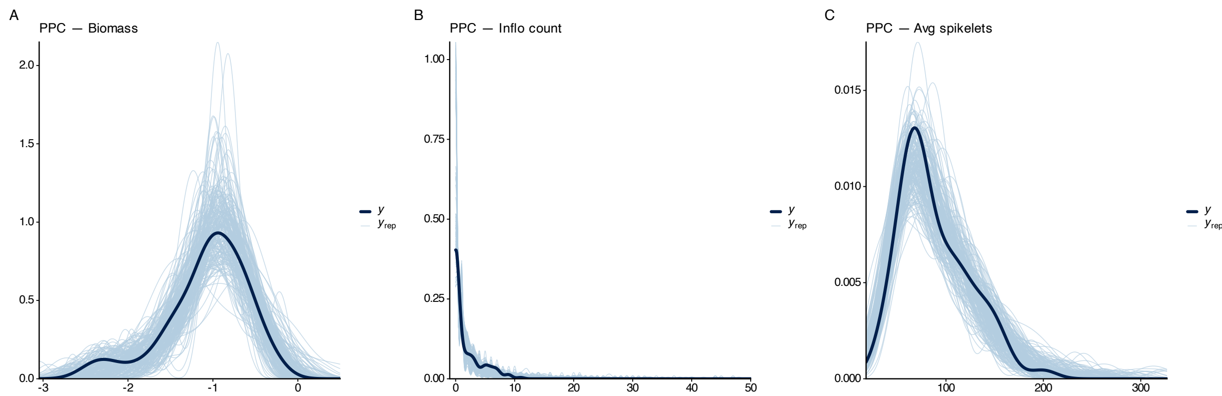

Fig S6: Posterior predictive checks for Bayesian models of plant performance in the greenhouse experiment. Density overlays show observed data (solid line) and 200 posterior predictive draws (shaded lines) for (A) aboveground biomass, (B) inflorescence count, and (C) average spikelets per plant. Models adequately capture the distribution of observed data across all response variables.

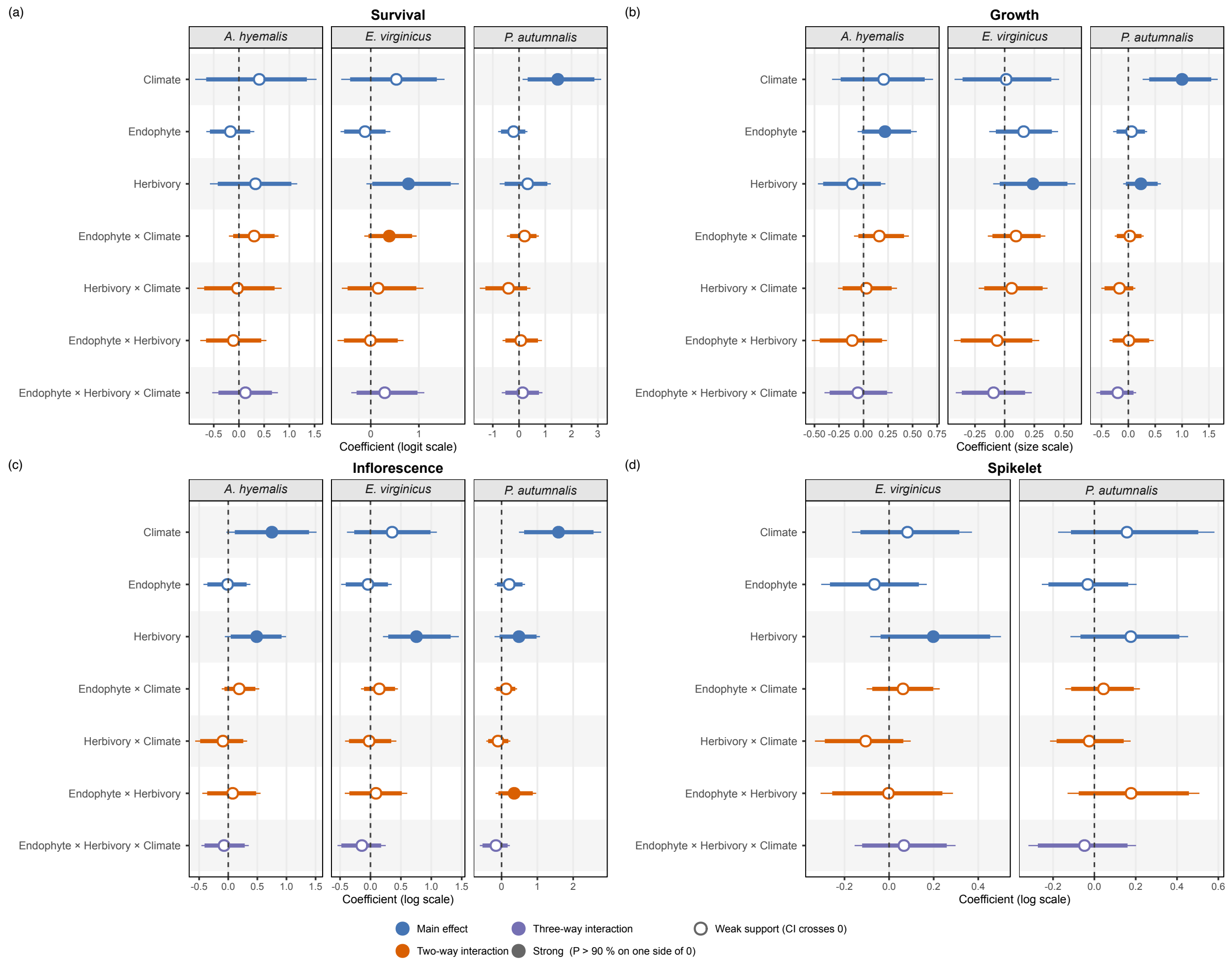

Fig S7: Posterior summaries of standardized regression coefficients from Bayesian models of survival, growth, inflorescence production, and spikelet production across three cool-season grass species (*Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis*; spikelet production was modeled for *E. virginicus* and *P. autumnalis* only). Panels show median posterior estimates (points), 90% credible intervals (thick bars), and 95% credible intervals (thin bars) for main effects of climate (precipitation), endophyte presence, and herbivory, as well as their two-way and three-way interactions. Filled points indicate strong posterior support for an effect ($\Pr(\Delta > 0)$ or $\Pr(\Delta < 0) > 0.90$), whereas open points indicate weaker support. Colors distinguish main effects, two-way interactions, and three-way interactions. Vertical dashed lines indicate no effect ($\beta = 0$). Coefficients are shown on the model link scale: logit for survival, linear size scale for growth, and log scale for inflorescence and spikelet production.

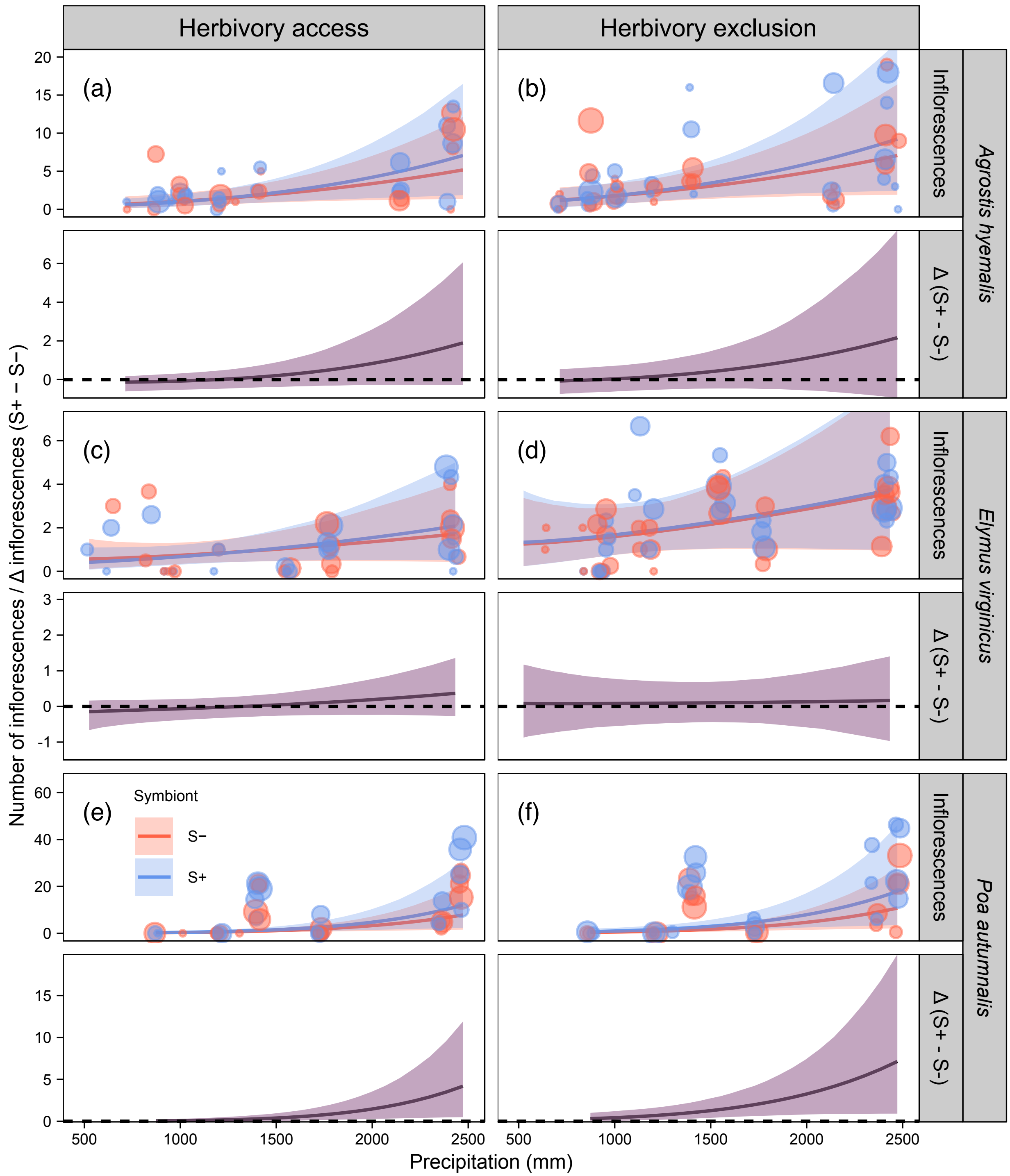

Fig S8: Predicted inflorescence production (upper panels) and symbiont effect (Δ = S+ − S-; lower panels) across the precipitation gradient. Lines show posterior means with 90% credible intervals. Points represent mean observed inflorescences per plot, slightly jittered for visualization. The dashed line at $\Delta = 0$ indicates no symbiont effect. Panels are arranged by species and herbivory treatment (herbivore access and exclusion). All panels share a common x-axis for illustrative purposes; some species extend beyond their observed precipitation range.

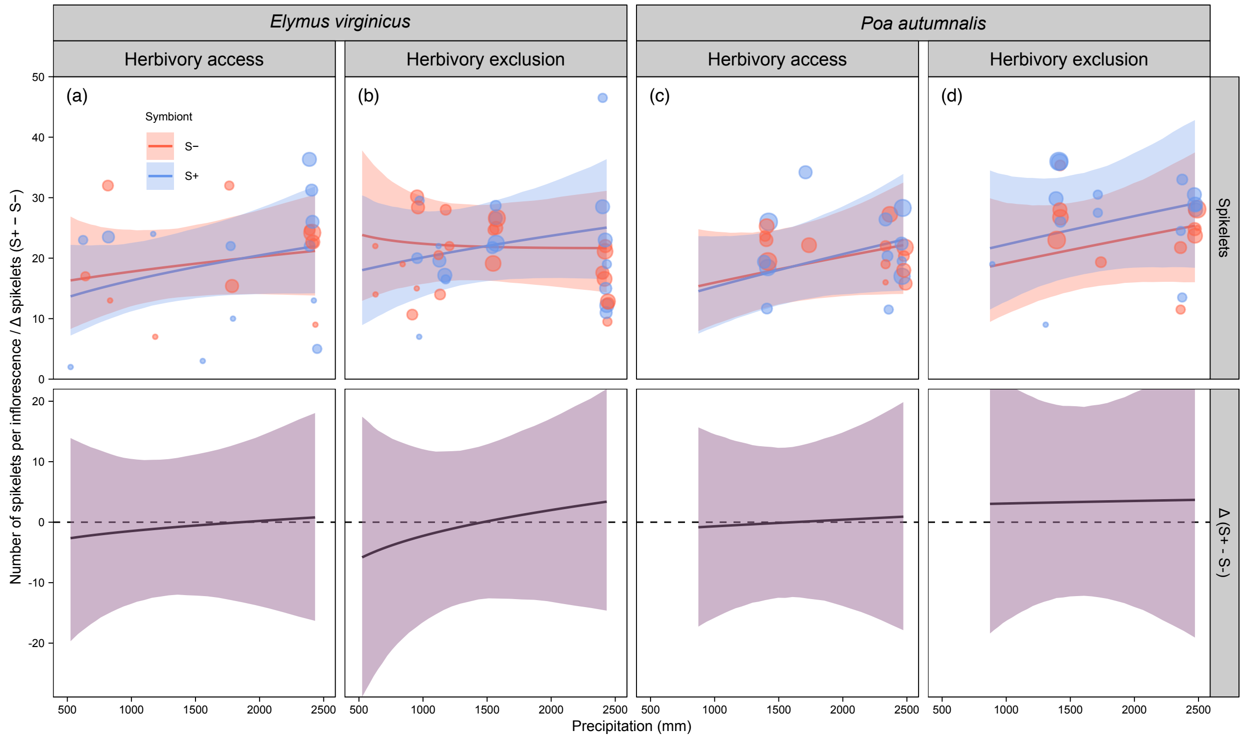

Fig S9: Predicted spikelet production (upper panels) and symbiont effect (Δ = S+ − S-; lower panels) across the precipitation gradient. Lines show posterior means with 90% credible intervals. Points represent mean observed spikelets per plot, slightly jittered for visualization. Spikelets were measured for two species only (see Methods). The dashed line at $\Delta = 0$ indicates no symbiont effect. Panels are arranged by species and herbivory treatment (herbivore access and exclusion). All panels share a common x-axis for illustrative purposes; some species extend beyond their observed precipitation range.

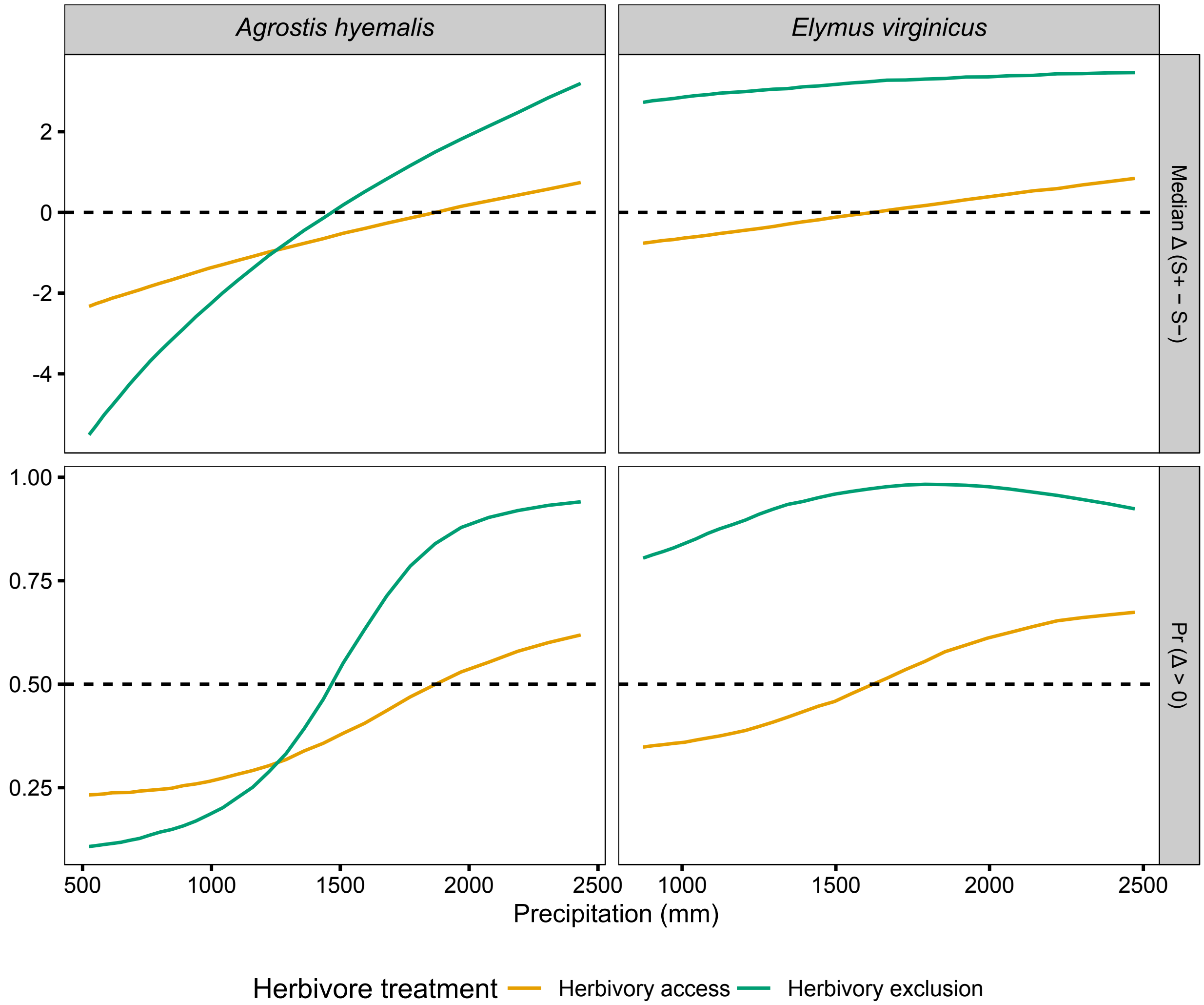

Fig S10: Median Δ (S+ − S-) and posterior probability $P(\Delta > 0)$ for spikelet production across precipitation quantiles for each species and herbivory treatment. Herbivore exclusion treatments are indicated by colour (Fenced = herbivores excluded; Unfenced = herbivores had access). Horizontal dashed lines indicate $\Delta = 0$ for medians and $P(\Delta > 0) = 0.5$ for probabilities. Panels are arranged by metric and species.